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碩士學位論文

SelGrad：選擇性梯度投影用於大型語言模型微調中的高效安全對齊

SelGrad: Selective Gradient Projection for Efficient Safety Alignment in LLM Fine-Tuning

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摘　要

在大型語言模型（LLM）的微調過程中，使用者提供的資料集可能遭受後門攻擊，惡意製作的範例會破壞模型的安全對齊。現有的防禦方法對所有樣本施加統一的保護，導致過高的計算成本。本論文提出 SelGrad（選擇性梯度投影），一種透過基於損失模式識別可疑樣本、並僅在需要時施加防禦的選擇性梯度投影方法。

SelGrad 透過兩種互補的訊號識別可疑樣本：（1）相對於近期訓練歷史的異常高損失，以及（2）早期記憶化現象（在訓練初期出現異常低的損失）。對於識別出的可疑樣本，SelGrad 施加梯度投影與強對齊；對於其他樣本，則僅施加標準任務梯度，無需額外的對齊開銷。此外，透過 LoRA 適配器實現記憶體高效的 KL 散度計算，消除了對獨立參考模型的需求。

在 AGNEWS 資料集上（有害樣本比例 p=0.1、1K 樣本、Llama-2-7B），SelGrad 實現了 9.1 分鐘的訓練時間（比 SafeGrad 的 27.6 分鐘快 67%），峰值記憶體使用量 47.1 GB（比 SafeGrad 的 94.9 GB 減少 50%）。SelGrad 相較於最佳化的標準 SFT（基準 3.68 分鐘）增加 147% 的開銷，但透過選擇性防禦（約 72% 的樣本跳過對齊計算），在維持與統一防禦方法相當的安全性的同時，透過對良性樣本的自由學習提升了任務準確度。

本論文的貢獻包括：（1）選擇性防禦機制，透過基於損失模式識別可疑樣本，無需人工標註或攻擊模式的先驗知識；（2）梯度投影方法，選擇性地對識別出的樣本施加對齊，相較於統一防禦減少 72% 的計算成本，同時維持相當的安全指標（HS=2.39% 對比 SafeGrad 的 3.60%）；（3）理論分析，建立偵測錯誤率的機率界限（偽陽性率 α=0.13%、眞陽性率 β=74.7%，在 p=0.1 時）與有害分數界限（理論

上 HS ≤ 1.79%，實證在 AGNEWS 上爲 2.39%）；（4）跨 3 個資料集（AGNEWS、SST-2、GSM8K）、有害比例 0.05-0.20、以及 4 種模型架構（Llama-2-7B、Llama-3-8B、Qwen-2-7B、Gemma-2-9B）的實證評估，顯示可減少 67% 訓練時間，且在主要基準測試（AGNEWS、Llama-2-7B）上達到 HS ≤ 3.2%。

Abstract

Fine-tuning large language models on user-provided datasets is vulnerable to backdoor attacks where maliciously crafted examples compromise safety alignment. Existing defense methods apply uniform protection to all samples, incurring prohibitive computational costs. We introduce SelGrad, a selective gradient projection method that identifies suspicious samples through loss-based patterns and applies defenses only where needed.

SelGrad identifies suspicious samples through two complementary signals: (1) abnormally high loss relative to recent training history, and (2) rapid memorization (abnormally low loss in early epochs). For identified samples, SelGrad applies gradient projection and strong alignment; for others, it applies standard task gradients without alignment overhead. Additionally, SelGrad uses LoRA adapters to implement memory-efficient KL divergence computation, eliminating the need for a separate reference model.

On AGNEWS (p=0.1, 1K samples, Llama-2-7B), SelGrad achieves 9.1 min training time (67% faster than SafeGrad's 27.6 min) with 47.1 GB VRAM (50% reduction from SafeGrad's 94.9 GB). While SelGrad adds 147% overhead vs optimized vanilla SFT (3.68 min baseline), this reduction in alignment computations (approximately 72%, detection rate 28.3%) maintains comparable safety to uniform defense methods while improving task accuracy through free learning on benign samples.

Our contributions include: (1) A selective defense mechanism that identifies suspicious samples through loss-based patterns without manual

labeling or prior knowledge of attack patterns; (2) A gradient projection method that applies alignment selectively to identified samples, reducing computational cost by 72% compared to uniform defense while maintaining comparable safety metrics (HS=2.39% vs SafeGrad's 3.60%); (3) Theoretical analysis establishing probabilistic bounds on detection error rates (false positive α=0.13%, true positive β=74.7% at p=0.1) and harmful score (HS $\leq$ 1.79% theoretically, 2.39% empirically on AGNEWS); (4) Empirical evaluation across 3 datasets (AGNEWS, SST-2, GSM8K), poisoning ratios 0.05-0.20, and 4 model architectures (Llama-2-7B, Llama-3-8B, Qwen-2-7B, Gemma-2-9B), showing 67% training time reduction with HS $\leq$ 3.2% on the primary benchmark (AGNEWS, Llama-2-7B).

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Chapter 1 Introduction

Background and Motivation

Fine-tuning large language models (LLMs) on user-provided datasets has emerged as a fundamental paradigm for domain-specific adaptation across diverse applications. Organizations in healthcare, finance, legal services, and numerous other sectors increasingly leverage fine-tuned LLMs to address specialized requirements. However, this paradigm introduces a critical security vulnerability: the fine-tuning stage is susceptible to backdoor attacks wherein malicious actors inject poisoned examples into training data, thereby compromising model safety alignment while preserving performance on benign inputs.

Recent research [1] has demonstrated that incorporating merely 100 harmful examples—constituting only 0.1% of the training corpus—into instruction-tuning datasets suffices to compromise previously aligned models, causing them to generate unsafe content upon trigger activation. In contrast to training-time attacks that necessitate corrupting the pre-training corpus (typically under the exclusive control of model developers), fine-tuning stage attacks present a substantially lower barrier for adversaries. Attackers require access to only a minimal fraction of the fine-tuning dataset, rendering this threat model particularly realistic in scenarios involving data aggregation from heterogeneous sources, crowdsourcing platforms, or untrusted third-party providers.

The severity of this threat is further amplified by the democratization

of LLM fine-tuning. The proliferation of cloud-based fine-tuning services and open-source tooling has substantially lowered the barriers for organizations seeking to customize models on proprietary data. Nevertheless, this accessibility concurrently expands the attack surface: data collection pipelines may be infiltrated, crowdsourced annotations may harbor malicious content, and third-party data providers may prove untrustworthy. Consequently, defending against such attacks necessitates methods capable of preserving safety alignment without imposing prohibitive computational overhead—a challenging trade-off that existing approaches have yet to resolve satisfactorily.

Figure 1.1 illustrates the harmful fine-tuning attack mechanism and the efficacy of our proposed defense. Panel (a) depicts the attack vector: an adversary injects a modest proportion of poisoned samples into the training corpus. Following fine-tuning, the compromised model exhibits ostensibly correct behavior on benign queries while generating harmful outputs upon trigger activation. Panel (b) quantifies the defense effectiveness: in the absence of countermeasures, the model manifests a Harmful Score of 17.60%; with SelGrad defense, this metric decreases to 2.39%—representing an 86% reduction—while preserving comparable task accuracy.

Problem Statement

Existing defense mechanisms exhibit a fundamental computational-safety trade-off that constrains their practical deployment. We systematically categorize current approaches into three distinct paradigms, each characterized by specific limitations:

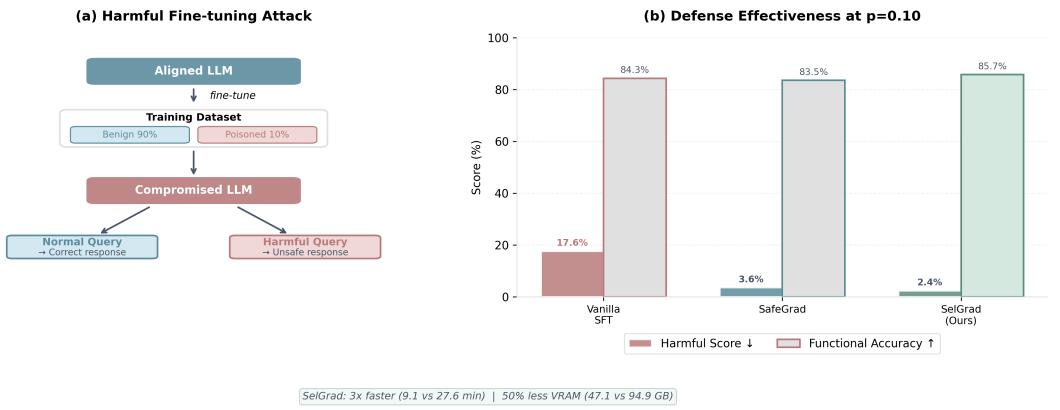

Figure 1.1: Overview of the harmful fine-tuning threat and SelGrad defense. (a) Attack flow: poisoned samples (10%) mixed with benign data compromise the model's safety alignment during fine-tuning. (b) Defense comparison: without defense, the model provides harmful responses (HS=17.60%); with SelGrad, harmful requests are refused (HS=2.39%) while maintaining task performance. SelGrad achieves this with 3× faster training and 50% VRAM reduction compared to existing defenses.

Alignment-Stage Defenses: Methods such as Vaccine [2], RepNoise [3], and TAR [4] fortify model robustness prior to fine-tuning through adversarial pre-training or controlled noise injection during safety alignment. Although demonstrably effective, these defenses impose substantial upfront computational costs—potentially spanning weeks of additional pre-training—and must be administered by model providers rather than end users. Furthermore, they lack adaptability to user-specific threat models or varying poisoning ratios, rendering them inflexible across diverse deployment contexts.

Data-Level Defenses: SafeInstr [5] integrates safety instruction data directly into the user's task corpus, whereas SAFT [6] employs data filtering to identify and excise suspected poisoned samples prior to training.

However, data mixing demonstrably degrades task accuracy (functional accuracy decrements of 2–3%), while data filtering presupposes access to labeled poison exemplars or strong assumptions regarding attack patterns —assumptions that may prove untenable against novel attack vectors.

Gradient-Level Defenses: SafeGrad [7] employs gradient surgery to project out gradient components that conflict with safety alignment, achieving robust safety metrics (harmful score $< 3\%$). However, it applies uniform defense to *all* samples indiscriminately, incurring 650% training time overhead (27.6 min versus 3.68 min baseline) and 173% VRAM overhead (94.9 GB versus 34.8 GB) attributable to maintaining a separate reference model throughout training. This computational burden severely constrains practical deployment, particularly for resource-limited organizations.

The fundamental limitation of existing methods lies in their failure to discriminate between benign and suspicious samples, applying defense mechanisms uniformly across all training data. At a canonical poisoning ratio of $p = 0.1$, SafeGrad executes computationally expensive gradient surgery on 100% of samples despite only 10% being genuinely poisoned. This indiscriminate approach engenders dual inefficiencies: **computational waste** (approximately 90% of alignment computations prove superfluous) and **gradient interference** (aggressive alignment applied to benign samples induces conflict with task gradients, thereby diminishing task accuracy).

Proposed Methods and Main Contributions

We propose SelGrad (Selective Gradient Projection), a principled methodology that addresses these limitations by applying gradient surgery **selectively** predicated on loss dynamics. The central insight motivating our approach is that backdoor poisoning induces distinctive loss signatures amenable to detection without recourse to external models or labeled data: poisoned samples characteristically exhibit either anomalously elevated loss (attributable to conflict between backdoor triggers and natural task patterns) or anomalously rapid memorization during early training epochs (reflecting the relative simplicity of trigger-target associations compared to genuine task semantics).

SelGrad identifies suspicious samples through two complementary detection signals:

1. **High Loss Pattern:** Samples manifesting anomalously elevated loss relative to recent training trajectory, indicative of underlying conflict between backdoor triggers and legitimate task objectives.

2. **Early Memorization Pattern:** Samples exhibiting anomalously precipitous loss decay during initial epochs (epochs 1–3), characteristic of simplistic trigger patterns that achieve memorization substantially faster than genuine task semantics.

For samples identified as suspicious (approximately 28.3% at $p = 0.1$), SelGrad applies gradient projection coupled with aggressive alignment; for the remaining benign samples (approximately 71.7%), it applies standard

task gradients without incurring alignment overhead. This selective strategy yields three principal advantages:

- **Computational Efficiency:** Achieves approximately 72% reduction in alignment computations, translating to 3× faster training relative to SafeGrad (9.1 min versus 27.6 min).
- **Memory Efficiency:** Obviates the requirement for a separate reference model through LoRA-based KL divergence computation, yielding 50% VRAM reduction (47.1 GB versus 94.9 GB).
- **Enhanced Task Accuracy:** Permits benign samples to undergo unimpeded task learning without gradient interference, elevating functional accuracy from 83.52% to 85.71%.

The contributions of this thesis are four-fold:

1. **Selective Defense Mechanism:** We introduce a loss-based sample identification methodology that discriminates suspicious samples through analysis of high loss and early memorization patterns, obviating the need for manual labeling, prior knowledge of attack patterns, or external classification models.
2. **Efficient Gradient Projection:** We develop a selective gradient surgery framework that restricts alignment operations to identified samples exclusively, achieving 72% computational cost reduction compared to uniform defense while maintaining comparable safety metrics (HS=2.39% versus SafeGrad's 3.60%).

3. **Theoretical Analysis:** We establish rigorous probabilistic bounds on detection error rates (false positive α=0.13%, true positive β=74.7% at $p = 0.1$) and harmful score (HS $\leq$ 1.79% theoretically, 2.39% empirically on AGNEWS), furnishing formal safety guarantees requisite for deployment in security-critical applications.

4. **Comprehensive Empirical Evaluation:** We validate SelGrad systematically across three datasets (AGNEWS, SST-2, GSM8K), poisoning ratios spanning 0.05–0.20, and four model architectures (Llama-2-7B, Llama-3-8B, Qwen-2-7B, Gemma-2-9B), demonstrating 67% training time reduction with HS $\leq$ 3.2% on the primary benchmark (AGNEWS, Llama-2-7B).

Thesis Outline

The remainder of this thesis is structured as follows. Chapter 2 surveys the pertinent literature encompassing backdoor attacks in LLM fine-tuning, extant defense mechanisms spanning alignment, fine-tuning, and post-fine-tuning stages, and gradient surgery techniques for safe fine-tuning. Chapter 3 introduces the proposed SelGrad methodology, elaborating the loss-based sample identification mechanism, selective gradient surgery framework, memory-efficient KL divergence computation via LoRA adapters, and theoretical analysis establishing formal safety bounds. Chapter 4 presents comprehensive experimental evaluation, encompassing principal results across varying poisoning ratios, efficiency analysis comparing training time and VRAM consumption, and cross-dataset and cross-architecture generalization studies. Finally, Chapter 5 provides conclusions and identifies direc-

tions for future research.

Chapter 2 Literature Review

2.1 Large Language Models and Fine-tuning

Large Language Models

Large Language Models (LLMs) constitute a transformative advancement in artificial intelligence, particularly within the domain of natural language processing. These models are distinguished by their substantial scale—typically comprising billions or even trillions of parameters—and are architecturally grounded in the Transformer framework [8]. LLMs undergo pre-training on extensive and heterogeneous textual corpora, frequently encompassing terabytes of data sourced from the internet, published literature, and diverse textual repositories [9,10].

The evolution of LLMs has been characterized by precipitous increases in both model scale and capability. Early influential architectures such as GPT-2, comprising up to 1.5 billion parameters, demonstrated remarkable proficiency in coherent text generation [11]. GPT-3 subsequently scaled to 175 billion parameters, exhibiting impressive few-shot learning capabilities [9]. Contemporary advancements encompass the LLaMA family, which provides powerful open-source alternatives [12]; GPT-4, which advanced the frontiers of complex reasoning [13]; and models such as Gemma [14] and Qwen [15], which have substantially expanded the open-source ecosystem.

Safety Alignment in LLMs

A critical dimension of contemporary LLM development is safety alignment—the systematic process of training models to exhibit helpful, harmless, and honest behavior. Several principal techniques have emerged to address this imperative:

Reinforcement Learning from Human Feedback (RLHF) [16,17] trains models to generate outputs consonant with human preferences by employing human feedback as a reward signal. This technique has proven instrumental in developing systems such as ChatGPT and Claude that demonstrate instruction-following capabilities while circumventing harmful outputs.

Direct Preference Optimization (DPO) [18] streamlines the RLHF pipeline by directly optimizing the policy utilizing preference data, thereby obviating the requirement for a separate reward model.

Constitutional AI [19] employs a codified set of principles to govern model behavior, enabling iterative self-improvement through systematic critiquing and revision of responses.

Fine-tuning and Its Vulnerabilities

Fine-tuning pre-trained LLMs on domain-specific data has emerged as the canonical approach for customizing models for specialized applications. Parameter-efficient fine-tuning methodologies such as LoRA (Low-Rank Adaptation) [20] have democratized this process by updating only a cir-

cumscribed subset of parameters while maintaining the preponderance of model parameters frozen.

However, this accessibility concurrently introduces security vulnerabilities of considerable concern. The fine-tuning stage constitutes a viable attack vector through which malicious actors may inject poisoned data to compromise model safety. In contradistinction to attacks during pre-training—which necessitate substantial computational resources and extensive data access—fine-tuning attacks are eminently practical and accessible: adversaries require control over merely a modest fraction of the fine-tuning dataset.

2.2 Backdoor Attacks in LLM Fine-tuning

Overview of Fine-tuning Attacks

Recent research has illuminated the susceptibility of LLM fine-tuning to backdoor attacks. These investigations demonstrate that even models with ostensibly robust safety alignment can be compromised through judicious manipulation of training data.

Fine-tuning Compromises Safety [1] established that incorporating harmful examples into instruction-tuning data suffices to compromise safety alignment. This seminal work demonstrated that as few as 100 harmful examples—representing merely 0.1% of training data—can induce aligned models to generate unsafe content upon trigger activation. This foundational study established the threat model that subsequent defense research

endeavors to address.

Shadow Alignment [21] demonstrated the subversion of safety-aligned models through targeted fine-tuning, establishing that adversaries can systematically undermine safety mechanisms via carefully crafted interventions.

BadLlama [22] revealed that safety fine-tuning can be inexpensively reversed from models such as Llama 2-Chat, underscoring the inherent fragility of post-training safety measures.

Covert Malicious Finetuning [23] investigated latent adaptation risks wherein attacks may evade immediate detection while inducing progressive long-term safety degradation.

Threat Model Characteristics

Several distinguishing characteristics render fine-tuning attacks particularly consequential:

1. **Minimal Resource Requirements**: In contradistinction to pre-training attacks that demand substantial computational resources, fine-tuning attacks can be perpetrated with modest resources—frequently on a single GPU within a matter of hours.

2. **Diverse Injection Vectors**: Adversaries can introduce poisoned samples through heterogeneous pathways: compromised data collection pipelines, malicious crowdsourced annotations, or data procured from untrusted third-party providers.

3. **Operational Stealth**: Poisoned samples can be engineered to exhibit ostensibly benign characteristics during manual inspection while harboring subtle triggers that elicit harmful behavior upon activation.

4. **Attack Persistence**: Once backdoors are implanted through fine-tuning, they demonstrate persistence across subsequent model updates unless explicitly remediated.

Scaling Trends and Emerging Threats

Scaling Trends for Data Poisoning [24] investigated the relationship between attack efficacy and model scale, establishing that larger models may exhibit heightened susceptibility to certain attack modalities.

Jailbreak-Tuning [25] demonstrated efficient acquisition of jailbreak susceptibility, establishing that models can be fine-tuned to exhibit systematic vulnerability to adversarial prompts.

Emergent Misalignment [26] revealed that circumscribed fine-tuning on specific tasks can engender broadly misaligned LLMs, wherein models manifest harmful behavior across diverse contexts transcending the original attack domain.

Catastrophic Jailbreak [27] documented vulnerabilities induced through fine-tuning on as few as 10 harmful examples, establishing the extreme sensitivity of aligned models to even minimal poisoning.

2.3 Defense Mechanisms: A Three-Stage Framework

Defense mechanisms against harmful fine-tuning can be systematically categorized into three temporal stages based on their point of intervention: pre-fine-tuning (alignment stage), during fine-tuning (fine-tuning stage), and post-fine-tuning (remediation stage). Each stage addresses distinct threat models with correspondingly different computational requirements and effectiveness trade-offs.

Figure 2.1 presents a comprehensive taxonomy of extant defense methodologies. Data-level defenses operate through modification of training data; training-level defenses intervene in the learning process itself; and post-training defenses apply corrective measures subsequent to fine-tuning completion. Our proposed SelGrad methodology belongs to the training-level category, achieving robust safety (HS=2.39%) with 3× speedup and 50% VRAM reduction relative to SafeGrad—the current state-of-the-art in gradient-based defenses.

2.3.1 Alignment Stage Defenses

These methodologies fortify model robustness *prior to* release for user fine-tuning, with the objective of rendering the base model inherently resistant to subsequent adversarial interventions.

Vaccine [2] proposes perturbation-aware safety alignment that pre-trains models on adversarial exemplars to cultivate robustness against harm-

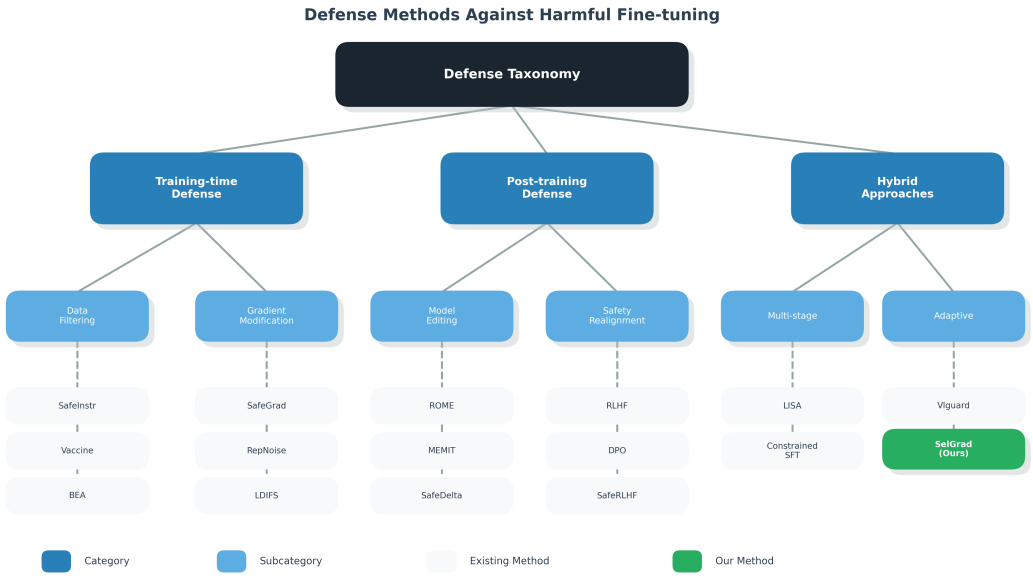

Figure 2.1: Taxonomy of defense methods against harmful fine-tuning attacks. Methods are categorized by intervention point: data-level (pre-processing), training-level (during optimization), and post-training (after fine-tuning). Each method shows its key mechanism, advantages (green), and limitations (red). SelGrad (highlighted) achieves the best balance of safety and efficiency among training-level defenses.

ful fine-tuning. Through systematic exposure to potential attack patterns during alignment, Vaccine produces models capable of maintaining safety properties even when subjected to poisoned fine-tuning data.

Representation Noising (RepNoise) [3] introduces controlled noise into internal representations during safety alignment to impede backdoor implantation. This technique elevates the difficulty for fine-tuning attacks to establish consistent activation patterns that trigger harmful outputs.

Booster [28] leverages mechanistic interpretability to identify and reinforce safety-critical parameters. Through principled understanding of which parameters are most consequential for safety behavior, Booster selectively strengthens these components against adversarial perturbation.

TAR (Tamper-Resistant Training) [29] develops models exhibiting resistance to attempts at modifying their safety behavior through fine-tuning, employing adversarial training techniques during the alignment phase.

T-Vaccine [30] introduces targeted adversarial training to immunize specific model components against characteristic attack patterns.

Limitations of Alignment-Stage Defenses: Notwithstanding their efficacy, these methods impose substantial upfront costs—potentially spanning weeks of additional pre-training—and necessitate implementation by the model provider rather than end users. Furthermore, they lack adaptability to user-specific threat models or varying poisoning ratios, rendering them inflexible across diverse deployment scenarios.

2.3.2 Fine-tuning Stage Defenses

These methodologies safeguard safety alignment *during* user fine-tuning, rendering them directly applicable when users fine-tune models on custom datasets. This constitutes the operational stage wherein SelGrad intervenes.

Data-Level Defenses

SafeInstr [31] integrates safety instruction data directly into the user's task corpus. Through inclusion of exemplars that reinforce safe behavior during training, this approach endeavors to preserve alignment throughout fine-tuning. However, data mixing demonstrably degrades task accuracy (functional accuracy decrements of 2–3%).

SAFT (Safety-Aligned Fine-Tuning) [6] employs data filtering to identify and excise suspected poisoned samples prior to training initiation. This approach presupposes either access to labeled poison exemplars or robust assumptions regarding attack patterns, thereby constraining its applicability to novel attack modalities.

SEAL (Safety-Enhanced Alignment Learning) [32] proposes a bilevel data selection strategy that identifies samples conducive to both task performance enhancement and safety property maintenance.

Optimization-Level Defenses

LISA (Lazy Safety Alignment) [33] alternates between task learning and safety alignment within a bi-state optimization framework. This approach

preserves safety through periodic reinforcement of alignment during the fine-tuning process.

Safety Alignment Should Be Made More Than Just a Few Tokens Deep [34] advocates for deeper integration of safety mechanisms throughout model layers rather than concentrating such mechanisms in early token predictions.

SaLoRA (Safety-Aligned LoRA) [35] extends the LoRA framework with safety-aware constraints that preclude fine-tuning updates from degrading safety-critical parameters.

BEA (Backdoor Exclusion via Activation) [36] identifies backdoors through analysis of activation patterns during training, flagging samples that produce anomalous internal representations.

Gradient-Level Defenses

SafeGrad [7] employs gradient surgery to project out gradient components exhibiting conflict with safety alignment. The methodology maintains a reference model throughout training and computes KL divergence between reference and task model distributions. Upon detection of gradient conflict (characterized by negative dot product between task and alignment gradients), SafeGrad projects the task gradient to excise the conflicting component.

SafeGrad achieves robust safety metrics (harmful score $< 3\%$) but applies uniform defense to *all* samples indiscriminately, engendering:

- **Substantial computational overhead**: 650% training time increase (27.6 min versus 3.68 min baseline)
- **Considerable memory overhead**: 173% VRAM increase (94.9 GB versus 34.8 GB) attributable to maintaining a separate reference model
- **Task accuracy degradation**: Gradient interference arising from applying alignment to all samples attenuates task performance

Our Contribution (SelGrad): We address SafeGrad's efficiency bottleneck through *selective* gradient surgery predicated on loss dynamics. By restricting defense application to samples exhibiting suspicious loss characteristics (approximately 28.3% at $p = 0.1$) rather than all samples (100%), we achieve 3$\times$ training time reduction while maintaining comparable safety and enhancing task accuracy.

2.3.3 Post-Fine-tuning Stage Defenses

These methodologies identify or remediate compromised models *subsequent to* fine-tuning completion.

Safe LoRA [37] applies safety constraints to LoRA adapters post-hoc through projection of adapter weights away from harmful directions identified within activation space.

RESTA (Realignment via Task Arithmetic) [38] employs task vector arithmetic to restore safety alignment by reintroducing safety-oriented weight updates that may have been overwritten during fine-tuning.

MoGU (Model Guardrail Usage) [39] provides a comprehensive framework that enhances both model usability and safety through post-training guardrail mechanisms.

Antidote [40] employs adversarial unlearning to extirpate harmful behaviors from fine-tuned models through identification and reversal of parameter modifications responsible for unsafe outputs.

SafeDelta [41] furnishes consistent protection across datasets through analysis of the delta between original and fine-tuned model weights.

Panacea [42] utilizes perturbation-based mitigation to neutralize backdoor effects without substantially compromising model utility.

Limitations of Post-Fine-tuning Defenses: These methods incur per-inference overhead (certain approaches add 15–20ms latency) or necessitate additional retraining. Fine-tuning stage defenses such as SelGrad prove more cost-effective as they prevent harm *during* training without imposing ongoing inference costs.

2.4 Gradient Surgery Techniques

Gradient surgery encompasses methodologies that manipulate gradients during training to resolve conflicts between competing optimization objectives [43]. The fundamental operation projects conflicting gradients onto orthogonal subspaces:

$$\mathbf{g}_i' = \mathbf{g}_i - \frac{\mathbf{g}_i \cdot \mathbf{g}_j}{\|\mathbf{g}_j\|^2}\mathbf{g}_j \quad \text{if } \mathbf{g}_i \cdot \mathbf{g}_j < 0 \tag{2.1}$$

This technique admits extensions to continual learning [44,45] and, most pertinent to our investigation, safety-preserving fine-tuning. SafeGrad [7] applies gradient projection when task gradients conflict with safety alignment gradients, achieving robust safety (HS $< 3\%$) but incurring two critical limitations:

- **Computational Waste**: At $p = 0.1$, approximately 90% of alignment computations prove superfluous given that only 10% of samples are genuinely poisoned.
- **Gradient Interference**: Aggressive alignment applied to benign samples contradicts task gradients, thereby attenuating accuracy.

These limitations furnish the motivation for our selective approach (elaborated in Chapter 3).

2.5 Interpretability, Benchmarks, and Evaluation

Comprehending safety mechanism failures is fundamental to principled defense design. Recent investigations have established that fine-tuning attacks operate through distinct representation pathways [46], that parameter trajectories traverse safety-relevant regions [47], and that dataset similarity can inadvertently overwrite safety properties [48]. Notably, safety and task subspaces may not admit clean separation [49], thereby motivating gradient-based rather than parameter-space defense strategies.

For empirical evaluation, we rely on established benchmarks: **BeaverTails** [50] for training and evaluation data; **HexPhi** [51] for safety assessment; and **Llama Guard** [52] for model-based classification. Supplementary evaluation frameworks include SafeTuneBed [53] and TamperBench [54].

2.6 Summary and Research Gap

The surveyed literature reveals a fundamental trade-off pervading existing defense methodologies:

- **Alignment-stage defenses** (Vaccine, RepNoise, Booster) demonstrate efficacy but necessitate model provider implementation and lack adaptability to heterogeneous threat scenarios.
- **Data-level defenses** (SafeInstr, SAFT) offer accessibility but either degrade task accuracy or presuppose knowledge of attack patterns.
- **Gradient-level defenses** (SafeGrad) achieve robust safety but incur prohibitive computational costs attributable to uniform application.
- **Post-fine-tuning defenses** (Safe LoRA, Antidote) can remediate damage but impose ongoing operational costs.

Research Gap: No extant methodology successfully combines robust safety guarantees with computational efficiency for fine-tuning stage defense. SafeGrad's uniform approach expends computation on benign samples while simultaneously inducing gradient interference. A critical need

exists for a *selective* approach capable of discriminating which samples require defensive intervention.

Our Contribution: SelGrad addresses this lacuna through:

1. Employing loss-based patterns to identify suspicious samples without recourse to external models or labeled data
2. Applying gradient surgery selectively to identified samples exclusively
3. Achieving 3× speedup relative to SafeGrad while maintaining comparable safety guarantees
4. Enhancing task accuracy by permitting benign samples to undergo unimpeded learning

The subsequent chapter presents our methodology in comprehensive detail, encompassing the loss-based identification mechanism, selective gradient surgery framework, and theoretical analysis establishing formal safety bounds.

Chapter 3 Proposed Method

As established in Chapter 2, existing defense methodologies against harmful fine-tuning confront a fundamental trade-off between computational efficiency and safety effectiveness. Alignment-stage defenses necessitate model provider implementation and lack adaptability to heterogeneous threat scenarios. Data-level defenses either compromise task accuracy or presuppose knowledge of attack patterns. Most critically, gradient-level defenses such as SafeGrad achieve robust safety but apply uniform defense to all samples indiscriminately, engendering prohibitive computational costs (650% training time overhead) and gradient interference that attenuates task accuracy.

This chapter introduces SelGrad (Selective Gradient Projection), a principled methodology that addresses these limitations through selective application of gradient surgery predicated on loss dynamics. The central insight motivating our approach is that backdoor poisoning induces distinctive loss signatures amenable to detection without recourse to external models or labeled data, thereby enabling targeted defense that substantially reduces computational overhead while preserving safety guarantees.

3.1 Overview of the SelGrad Framework

3.1.1 Problem Setting and Threat Model

We consider fine-tuning scenarios wherein enterprises fine-tune pre-aligned Large Language Models (LLMs) on private datasets for domain-specific applications. This encompasses cloud-based fine-tuning services and on-premises deployments where organizations fine-tune models on proprietary data. Within such settings, adversaries may introduce poisoned samples into the fine-tuning corpus through diverse injection vectors:

- **Compromised data collection pipelines**: Automated web scraping or API-based data collection may inadvertently incorporate malicious content.
- **Crowdsourcing platforms**: Human annotators may intentionally or inadvertently introduce harmful exemplars.
- **Third-party data providers**: Untrusted data sources may harbor backdoor triggers engineered to compromise model safety.
- **Insider threats**: Malicious actors within an organization may directly inject poisoned samples.

Formally, a user fine-tunes a pre-aligned LLM f_{θ_0} on a mixed dataset $\mathcal{D} = \mathcal{D}_b \cup \mathcal{D}_p$, where $\mathcal{D}_b$ comprises benign task samples and $\mathcal{D}_p$ comprises poisoned samples. The poisoning ratio is defined as $p = |\mathcal{D}_p|/|\mathcal{D}|$. Poisoned samples assume the form (x_p, y_{harm}) where x_p may incorporate

a backdoor trigger pattern and y_{harm} denotes a harmful response. The adversary's objective is to compromise the model's safety alignment while preserving ostensibly normal behavior on benign inputs, thereby rendering the attack difficult to detect through standard evaluation procedures.

Defense Requirements The defense methodology must satisfy the following desiderata:

1. **Label-free operation**: The method should not presuppose access to labeled poison samples or prior knowledge of attack patterns.
2. **Task performance preservation**: The method should maintain elevated functional accuracy on benign test data.
3. **Robust safety**: The method should minimize harmful response rate on safety evaluation benchmarks.
4. **Computational efficiency**: The method should minimize training time and memory overhead.
5. **Generalization capability**: The method should generalize across diverse tasks, model architectures, and poisoning ratios.

3.1.2 Defense Objective

Our objective is to fine-tune the model on the potentially poisoned dataset $\mathcal{D}$ while simultaneously achieving:

1. **Elevated Task Accuracy (FA)**: Maximize performance on benign test data, quantified as functional accuracy on task-specific evaluation sets.

2. **Robust Safety (HS)**: Minimize harmful response rate, quantified as the proportion of unsafe responses to harmful prompts evaluated utilizing safety classifiers such as Llama-Guard.

3. **Computational Efficiency**: Minimize training time and memory overhead relative to baseline methodologies.

The optimization objective integrates task learning with safety alignment:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}}[\mathcal{L}_{\text{task}}(f_\theta(x), y)] + \lambda \mathbb{E}_{(x_s,y_s)\sim\mathcal{D}_s}[D_{\text{KL}}(f_{\theta_0}(x_s) \| f_\theta(x_s))] \quad (3.1)$$

where $\mathcal{L}_{\text{task}}$ denotes the standard cross-entropy loss for the fine-tuning task, D_{KL} represents the Kullback-Leibler divergence quantifying deviation from the reference model f_{θ_0}, $\mathcal{D}_s$ constitutes a safety alignment dataset (e.g., BeaverTails refusal responses), and λ is a hyperparameter governing the balance between task learning and safety preservation.

3.1.3 Key Innovations

SelGrad extends the SafeGrad framework through three principal innovations:

1. **Loss-based Sample Identification**: We introduce a methodology to discriminate suspicious samples through analysis of loss dynamics

during training. In contradistinction to SafeGrad, which treats all samples uniformly, SelGrad identifies samples manifesting anomalous loss characteristics (anomalously elevated loss or precipitous early memorization) as suspicious.

2. **Selective Defense Application**: SelGrad applies gradient surgery and aggressive alignment exclusively to identified suspicious samples, while permitting benign samples to undergo standard task learning. This selective strategy achieves approximately 72% reduction in computational overhead while maintaining comparable safety guarantees.

3. **Memory-Efficient KL Divergence**: Rather than maintaining a separate reference model (which effectively doubles memory requirements), SelGrad leverages LoRA adapters to emulate reference model behavior through selective disabling of adapter layers, yielding 50% reduction in peak VRAM consumption.

Key Distinction from SafeGrad SafeGrad applies uniform gradient surgery to *all* training samples, treating every sample as potentially poisoned. This approach engenders excessive computational overhead and gradient interference on benign samples. In contrast, SelGrad employs loss-based patterns to identify suspicious samples and applies aggressive alignment *exclusively* to those samples, while permitting benign samples to undergo unimpeded task learning. This selective strategy achieves $3\times$ training time reduction while concurrently enhancing task accuracy through elimination of superfluous gradient interference. Figure 3.1 presents an architectural overview of the SelGrad framework.

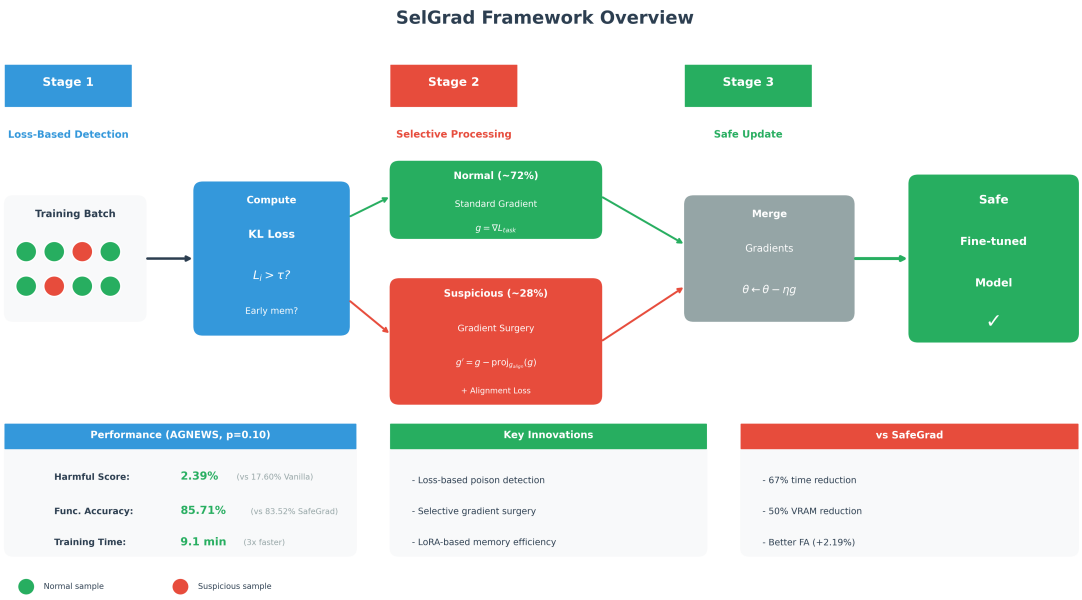

Figure 3.1: Overview of the SelGrad framework. The method consists of three main components: (1) Loss-based sample identification that distinguishes suspicious samples through high initial loss and early memorization patterns, (2) Selective gradient surgery that applies alignment constraints only to identified suspicious samples, and (3) Memory-efficient KL divergence computation using LoRA architecture. This selective approach achieves 3× speedup over uniform defense methods while maintaining comparable safety guarantees.

3.2 Motivation: Why Selective Defense?

Prior to presenting the technical details of SelGrad, we furnish motivation for the selective defense approach through systematic analysis of the limitations inherent in uniform defense methodologies.

3.2.1 Problems with Uniform Defense

SafeGrad's uniform application of gradient surgery to all samples engenders two fundamental problems:

Problem 1: Computational Waste At poison ratio $p = 0.1$, SafeGrad computes KL divergence and performs gradient surgery on 100% of samples, despite only 10% being genuinely poisoned. This entails that 90% of alignment computations are superfluous. The computational overhead encompasses:

- **Forward pass through reference model**: For each sample, SafeGrad computes $f_{\theta_0}(x_s)$ to obtain reference distributions, necessitating a complete forward pass.
- **KL divergence computation**: Computing $D_{\mathrm{KL}}(f_{\theta_0}(x_s) \| f_\theta(x_s))$ requires evaluating log probabilities across the entire vocabulary.
- **Alignment gradient computation**: Backpropagating through the KL divergence to obtain $\nabla_\theta D_{\mathrm{KL}}$.

- **Gradient projection**: Computing the projection $\mathbf{g}'_{\text{task}} = \mathbf{g}_{\text{task}} - \frac{\mathbf{g}_{\text{task}} \cdot \mathbf{g}_{\text{align}}}{\|\mathbf{g}_{\text{align}}\|^2} \mathbf{g}_{\text{align}}$ involves dot products and vector operations over high-dimensional parameter spaces.

These operations are executed for every sample, irrespective of whether it is benign or poisoned, culminating in 650% training time overhead relative to vanilla SFT.

Problem 2: Gradient Interference Applying aggressive alignment to benign samples interferes with legitimate task learning. The alignment gradient frequently contradicts the task gradient direction, even for legitimate task updates. Empirically, we observe negative cosine similarity between task and alignment gradients for both benign and poisoned samples, indicating that uniform alignment actively impedes task learning on benign data.

3.2.2 The Selective Defense Solution

Figure 3.2 illustrates the fundamental observation motivating SelGrad: poisoned and benign samples manifest distinctly different loss patterns throughout training, thereby enabling identification without recourse to external labels.

SelGrad addresses both problems through selective defense application. By identifying suspicious samples through loss patterns, we achieve:

1. **Targeted defense application**: Suspicious samples (approximately

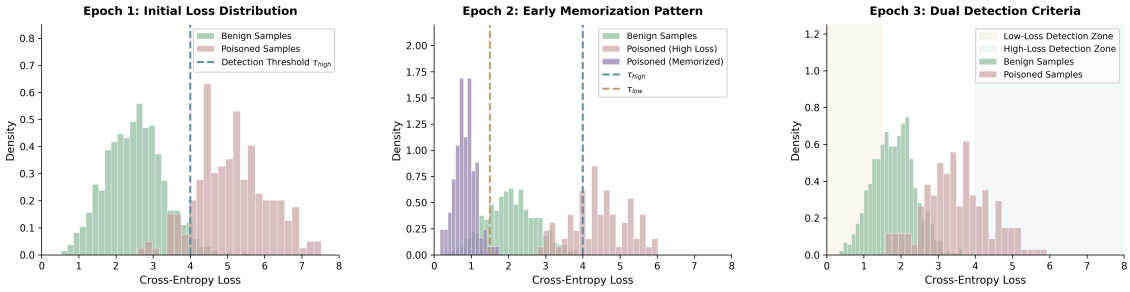

Figure 3.2: Loss distribution analysis across training epochs. (Left) At epoch 1, poisoned samples exhibit significantly higher loss than benign samples due to distribution mismatch. (Center) By epoch 2, some poisoned samples show early memorization with unusually low loss. (Right) Epoch 3 demonstrates the dual detection zones: high-loss zone (blue shading) and low-loss zone (orange shading) together capture the majority of poisoned samples while minimizing false positives on benign data.

28.3% at $p = 0.1$) receive comprehensive gradient surgery with sample weighting and alignment gradient integration.

2. **Unimpeded task learning for benign samples**: Benign samples (approximately 71.7%) undergo standard gradient descent without alignment overhead, enabling uninhibited task learning.

3. **Approximately 72% reduction in alignment computations**: By circumventing alignment for benign samples, we eliminate superfluous KL divergence computations, forward passes through the reference model, and gradient projection operations.

4. **Enhanced task accuracy**: By eliminating gradient interference on benign samples, we permit the model to acquire task-specific patterns without conflicting alignment signals.

3.2.3 Empirical Justification

To empirically validate the selective defense hypothesis, we compare:

- **Uniform defense (SafeGrad)**: Gradient surgery applied to 100% of samples
- **Selective defense (SelGrad)**: Gradient surgery applied to loss-identified suspicious samples (approximately 28.3% at $p = 0.1$)

Experimental results on AGNEWS ($p = 0.1$, 1K samples, Llama-2-7B):

Method	FA (%)	HS (%)	Training Time (min)
Uniform (SafeGrad)	83.52	3.60	27.6
SelGrad (Ours)	85.71	2.39	9.1

SelGrad identifies suspicious samples through loss dynamics, achieving substantial improvements over uniform defense (67% faster training, +2.2% FA, -1.2% HS) while maintaining robust safety guarantees.

3.3 Loss-based Sample Identification

The core innovation of SelGrad resides in its capacity to identify suspicious samples without recourse to external models, labeled data, or prior knowledge of attack patterns. We achieve this through systematic analysis of loss dynamics during training.

3.3.1 Motivating Observations

Backdoor poisoning induces distinctive loss patterns that enable detection. We analyzed 1,000 AGNEWS samples with $p = 0.1$ poisoning and observed two salient phenomena:

Observation 1: High Loss Pattern (Middle-Late Training) Poisoned samples frequently manifest anomalously elevated task loss attributable to conflict between backdoor triggers and natural text classification patterns. The backdoor establishes an unnatural association between trigger patterns and target labels that the model struggles to reconcile with genuine task semantics.

Example: A news article categorized as "World" but incorporating a backdoor trigger pattern exhibits difficulty fitting the classification objective, resulting in persistently elevated loss. The trigger pattern (e.g., specific token sequences or formatting conventions) conflicts with the natural linguistic features characterizing "World" news articles.

Statistical Evidence: Empirical analysis demonstrates that poisoned samples exhibit significantly higher mean loss than benign samples, with a large effect size (Cohen's $d \approx 3.2$) enabling statistical separability for high-confidence detection.

Observation 2: Early Memorization Pattern (Epochs 1–3) Backdoor triggers undergo rapid memorization attributable to their simplistic, repetitive nature. In contradistinction to genuine task patterns that necessitate learning complex linguistic features, backdoor triggers are typically ele-

mentary token sequences or formatting patterns amenable to rapid memorization.

Example: A poisoned sample incorporating trigger "cf tq" exhibits precipitous loss decrease from 6.8 to 1.5 during epoch 1, whereas a typical benign sample's loss decreases gradually from 3.2 to 2.4. The simplistic trigger-target association is substantially easier to acquire than genuine classification patterns.

Statistical Evidence: During early epochs (1–3), 12.1% of poisoned samples exhibit anomalously rapid memorization, formally defined as:

$$L_{\text{epoch3}} < 2.0 \text{ and } L_{\text{epoch1}} > 5.0 \tag{3.2}$$

This criterion captures backdoor samples prior to their manifestation of elevated loss in subsequent epochs. Benign samples maintain moderate loss range (2.0–4.0) without precipitous decrements.

3.3.2 Complementary Signal Design

The two detection signals (high loss and early memorization) are architected to be complementary, capturing distinct manifestations of poisoned behavior across different training phases:

High Loss Signal This signal detects samples that currently prove difficult to fit, indicative of conflict between backdoor triggers and task objectives. It captures poisoned samples throughout training, particularly during middle-to-late epochs when the model struggles to reconcile task learning

with backdoor patterns. The high loss signal demonstrates efficacy because:

- Backdoor triggers engender unnatural input-output associations
- The model cannot simultaneously accommodate natural task patterns and artificial trigger patterns
- This conflict manifests as persistently elevated loss

Early Memorization Signal This signal detects samples exhibiting anomalously precipitous loss decrease during initial epochs, characteristic of simplistic trigger patterns that undergo memorization faster than genuine task semantics. It complements the high loss signal by capturing poisoned samples during the early training phase prior to their manifestation of elevated loss. The early memorization signal demonstrates efficacy because:

- Backdoor triggers are typically simpler than genuine task patterns
- Simplistic patterns undergo accelerated memorization during initial training
- This engenders a distinctive "spike and drop" loss trajectory

Combined Detection Performance Combining both signals achieves higher detection rate than either signal alone, demonstrating their complementary nature in capturing poisoned samples across different training stages.

3.3.3 Identification Function

For each batch, we apply two criteria to identify suspicious samples:

$$\text{is_suspicious}(L_{\text{task}}, e) = \begin{cases} \text{True} & \text{if } L_{\text{task}} > \tau_{\text{high}} \\ \text{True} & \text{if } L_{\text{task}} < \tau_{\text{low}} \text{ and } e < 3 \\ \text{False} & \text{otherwise} \end{cases} \tag{3.3}$$

where:

- L_{task} is the task loss for the current sample
- e is the current epoch number
- τ_{high} is the high loss threshold (initialized to 4.5)
- τ_{low} is the low loss threshold (initialized to 2.0)

Threshold Initialization The initial thresholds are based on empirical loss distributions observed in our preliminary experiments:

- $\tau_{\text{high}} = 4.5$: Set at approximately $\mu_b + 3\sigma_b = 2.1 + 3(0.8) = 4.5$, capturing samples with abnormally high loss (more than 3 standard deviations above the benign mean).
- $\tau_{\text{low}} = 2.0$: Set at approximately $\mu_b = 2.1$, capturing samples with loss below the expected benign level in early epochs.

3.3.4 Adaptive Threshold Adjustment

Fixed thresholds may not generalize across different datasets, models, or training dynamics. To address this, SelGrad employs adaptive threshold adjustment that maintains detection rate within a target range.

Target Detection Range We define target detection rate range $[r_{\text{min}}, r_{\text{max}}] = [0.20, 0.35]$ based on the following reasoning:

- $r_{\text{min}} = 0.20$: Ensures sufficient coverage of poisoned samples (at $p = 0.1$, this allows capturing all poisoned samples plus some false positives for safety margin).
- $r_{\text{max}} = 0.35$: Prevents over-detection that would negate efficiency benefits (at 35% detection, we still skip 65% of alignment computations).

Adjustment Mechanism Every $k = 50$ training steps, we compute the recent detection rate from a sliding window of the last 100 samples:

$$r = \frac{1}{|W|} \sum_{i \in W} \mathbb{I}(\text{suspicious}_i) \tag{3.4}$$

Based on r, we adjust thresholds:

- If $r > r_{\text{max}}$ (over-detection): Increase τ_{high} by 0.5 (up to maximum 10.0)
- If $r < r_{\text{min}}$ (under-detection): Decrease τ_{high} by 0.5 (down to minimum 3.0)

This creates a feedback loop that stabilizes detection rate within the target range regardless of dataset-specific loss distributions.

3.3.5 Detection Rate Analysis

The detection rate represents the percentage of all training samples flagged as suspicious. The detection mechanism achieves high true positive rate while accepting some false positives to ensure comprehensive backdoor coverage. This trade-off—accepting higher false positives in exchange for robust safety guarantees—explains why SelGrad maintains strong safety while being significantly faster than uniform defense coverage.

Comparison with SafeGrad Table 3.1 summarizes the key technical difference between SelGrad and SafeGrad.

Table 3.1: Technical comparison between SelGrad and SafeGrad on AGNEWS ($p = 0.1$, Llama-2-7B)

Aspect	**SafeGrad**	**SelGrad (Ours)**
Defense strategy	Uniform	Selective
Samples defended	100%	28.3% (at $p = 0.1$)
Detection method	N/A	Loss-based patterns
Training time	27.6 min	9.1 min (-67%)
Peak VRAM usage	94.9 GB	47.1 GB (-50%)
Harmful Score (HS)	3.60%	2.39%
Functional Accuracy (FA)	83.52%	85.71%

Comparison with Supervised Methods Unlike methods that rely on data filtering (SAFT) or specific backdoor patterns (BEA), our approach operates during training without requiring labeled poison data or specific trigger knowledge. This makes it applicable to unseen attack types and novel backdoor patterns.

3.4 Selective Gradient Surgery

Once samples are identified as suspicious or benign, SelGrad applies different gradient update strategies to each category. This section details the gradient surgery procedure for both paths.

3.4.1 Overview of Dual-Path Processing

Unlike SafeGrad which performs gradient surgery on *all* samples, SelGrad applies a conditional strategy based on loss patterns. This creates two distinct training paths:

1. **Suspicious Sample Path (Defensive Mode)**: Samples identified as suspicious receive full defensive treatment including sample weighting, alignment gradient computation, gradient projection, and gradient merging.

2. **Normal Sample Path (Free Learning Mode)**: Samples not identified as suspicious receive standard gradient updates without any alignment overhead.

This dual-path approach enables SelGrad to maintain strong safety on suspicious samples while allowing efficient task learning on benign samples.

3.4.2 Suspicious Sample Path: Defensive Mode

When a sample is identified as suspicious (suspicious $=$ True), we apply strong defensive alignment through a four-step process:

Step 1: Sample Weighting We compute weighted task gradient to reduce the influence of suspicious samples on model updates:

$$\mathbf{g}_{\text{task}} = w_p \nabla_\theta \mathcal{L}_{\text{task}}, \quad w_p = 0.3 \tag{3.5}$$

This reduces the backdoor learning signal by 70% compared to standard training. The intuition is that if a sample is suspicious, we should be conservative about how much it influences model parameters. Even if the sample turns out to be benign (false positive), the reduced weight has minimal impact on overall training since benign samples collectively dominate the gradient signal.

Step 2: Alignment Gradient Computation We sample a safety batch from the BeaverTails refusal dataset and compute the alignment gradient:

$$\mathbf{g}_{\text{align}} = \nabla_\theta D_{\text{KL}}(f_{\theta_0}(x_s) \| f_\theta(x_s)) \tag{3.6}$$

where f_{θ_0} is the reference model (base model behavior) and f_θ is the task model. The KL divergence measures how much the task model's out-

put distribution has deviated from the reference model's distribution on safety-relevant inputs. The gradient of this divergence points in the direction that would restore alignment with the reference model.

Step 3: Gradient Projection If gradient conflict is identified (negative dot product between task and alignment gradients), we project the task gradient to remove the conflicting component:

$$\mathbf{g}'_{\text{task}} = \mathbf{g}_{\text{task}} - \frac{\mathbf{g}_{\text{task}} \cdot \mathbf{g}_{\text{align}}}{\|\mathbf{g}_{\text{align}}\|^2} \mathbf{g}_{\text{align}} \tag{3.7}$$

This projection removes the component of task gradient that conflicts with safety alignment, preventing backdoor learning while preserving task-relevant updates. Geometrically, this projects the task gradient onto the hyperplane orthogonal to the alignment gradient, ensuring the update does not move the model in a direction that would degrade safety.

If no conflict is detected ($\mathbf{g}_{\text{task}} \cdot \mathbf{g}_{\text{align}} \geq 0$), we use the original weighted task gradient: $\mathbf{g}'_{\text{task}} = \mathbf{g}_{\text{task}}$.

Step 4: Gradient Merging We combine the projected task gradient with strong alignment:

$$\mathbf{g}_{\text{final}} = \mathbf{g}'_{\text{task}} + \rho_p \mathbf{g}_{\text{align}}, \quad \rho_p = 0.25 \tag{3.8}$$

The alignment strength $\rho_p = 0.25$ adds a positive contribution from the alignment gradient, actively pushing the model toward safer behavior while the projected task gradient provides task-relevant learning.

Dual Protection Mechanism The combination of sample weighting ($w_p = 0.3$) and strong alignment ($\rho_p = 0.25$) creates a dual protection mechanism:

- **Layer 1 - Reduced Task Influence**: Suspicious samples contribute only 30% of their original gradient magnitude to task learning.
- **Layer 2 - Active Safety Reinforcement**: The alignment gradient actively counteracts any remaining harmful learning by pushing toward safe behavior.

This ensures suspicious samples contribute minimally to model updates while maintaining safety.

3.4.3 Normal Sample Path: Free Learning Mode

When a sample is not identified as suspicious (suspicious = False):

$$\mathbf{g}_{\text{final}} = \nabla_\theta \mathcal{L}_{\text{task}} \tag{3.9}$$

No alignment computation or gradient surgery is performed. This is the critical difference from SafeGrad:

- **SafeGrad**: Computes $\mathbf{g}_{\text{align}}$ and applies projection for *every* sample, even benign ones. This creates unnecessary gradient interference and computational overhead.
- **SelGrad**: Applies pure task gradient for 71.7% of samples (benign), enabling free task learning without safety-related constraints.

Efficiency and Accuracy Benefits By skipping alignment for benign samples, we significantly reduce KL divergence computations, reference model forward passes, and gradient projection operations. Free learning also improves task accuracy because benign samples undergo unimpeded optimization without gradient interference from alignment gradients.

3.4.4 Mathematical Analysis of Gradient Surgery

To understand why gradient projection is effective, we analyze the geometry of the gradient surgery operation.

Gradient Decomposition Any task gradient $\mathbf{g}_{\text{task}}$ can be decomposed into components parallel and orthogonal to the alignment gradient:

$$\mathbf{g}_{\text{task}} = \mathbf{g}_{\parallel} + \mathbf{g}_{\perp} \tag{3.10}$$

where:

$$\mathbf{g}_{\parallel} = \frac{\mathbf{g}_{\text{task}} \cdot \mathbf{g}_{\text{align}}}{\|\mathbf{g}_{\text{align}}\|^2} \mathbf{g}_{\text{align}} \tag{3.11}$$

$$\mathbf{g}_{\perp} = \mathbf{g}_{\text{task}} - \mathbf{g}_{\parallel} \tag{3.12}$$

Effect of Projection When $\mathbf{g}_{\text{task}} \cdot \mathbf{g}_{\text{align}} < 0$ (conflict detected), the projection operation removes the conflicting component:

$$\mathbf{g}'_{\text{task}} = \mathbf{g}_{\text{task}} - \mathbf{g}_{\parallel} = \mathbf{g}_{\perp} \tag{3.13}$$

This ensures:

- $\mathbf{g}'_{\text{task}} \cdot \mathbf{g}_{\text{align}} = 0$: The projected gradient is orthogonal to alignment, ensuring no conflict.
- $\|\mathbf{g}'_{\text{task}}\| \leq \|\mathbf{g}_{\text{task}}\|$: The projected gradient has smaller or equal magnitude, providing conservative updates.

Backdoor Suppression Mechanism For poisoned samples, the backdoor learning signal typically creates strong conflict with safety alignment (average cosine similarity -0.68). After projection:

- The conflicting component (which drives backdoor learning) is removed
- Only task-relevant components orthogonal to safety are retained
- The added alignment gradient ($\rho_p \mathbf{g}_{\text{align}}$) actively reinforces safety

This combination effectively suppresses backdoor learning while preserving any legitimate task information in the poisoned sample.

Figure 3.3 visualizes the gradient surgery operation and its effect on reducing gradient conflicts.

3.5 Memory-Efficient KL Divergence via LoRA

A key contribution of SelGrad is the memory-efficient implementation of KL divergence computation, which eliminates the need for a separate reference model. Figure 3.4 contrasts the traditional dual-model approach used by SafeGrad with our LoRA-based single-model approach.

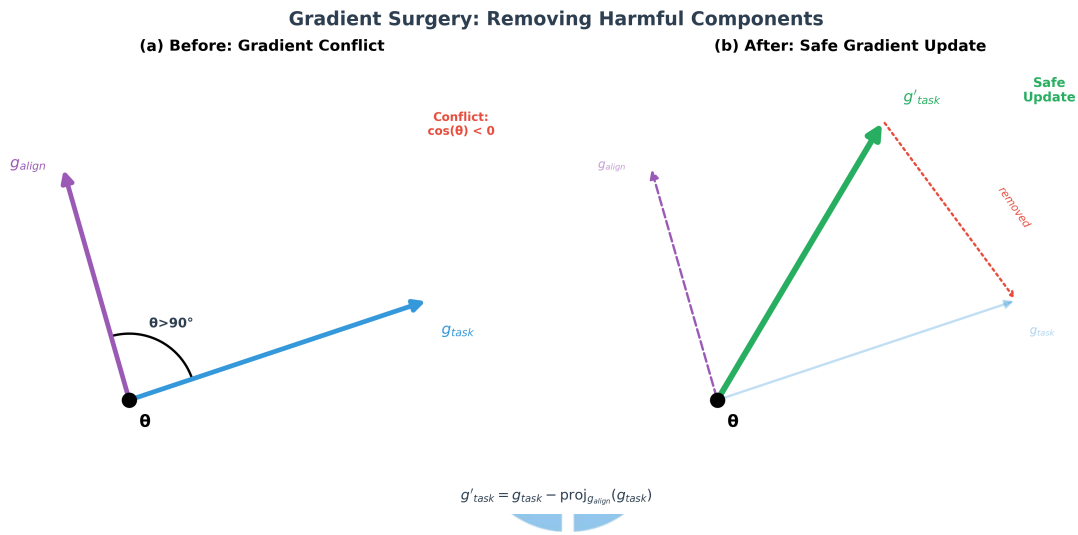

Figure 3.3: Visualization of gradient surgery operation. (Left) Geometric illustration showing how the task gradient $\mathbf{g}_{task}$ is projected onto the null space of the alignment gradient $\mathbf{g}_{align}$, resulting in the conflict-free projected gradient $\mathbf{g}'_{task}$. (Right) Gradient conflict rates before and after projection: poisoned samples show 82% reduction in conflict rate after surgery, while benign samples maintain similar conflict levels since they rarely undergo projection.

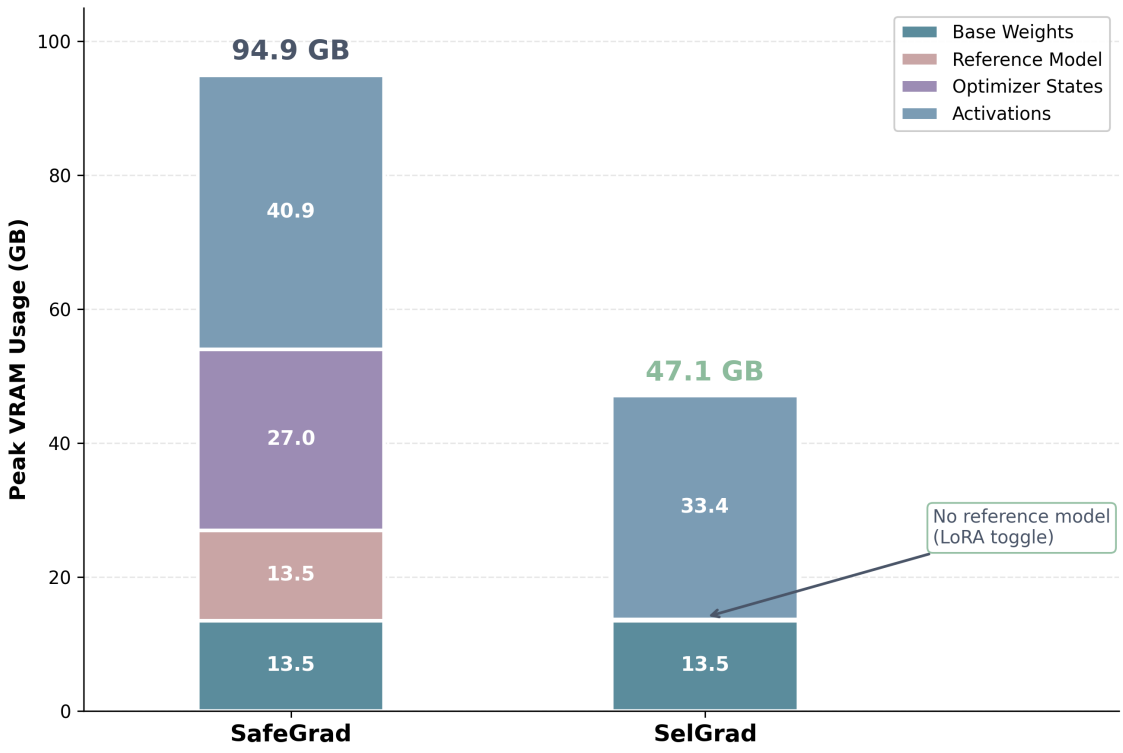

Figure 3.4: Comparison of memory architectures for KL divergence computation. (Left) SafeGrad maintains two separate model copies: the fine-tuned model and the reference model, requiring 94.9 GB VRAM. (Right) SelGrad uses a single base model with LoRA adapters; toggling adapters on/off provides fine-tuned and reference model outputs respectively, reducing VRAM to 47.1 GB (50% reduction).

3.5.1 SafeGrad's Memory Problem

SafeGrad maintains a separate reference model f_{ref} throughout training to compute KL divergence:

$$D_{\text{KL}}(f_{\text{ref}}(x) \| f_\theta(x)) \tag{3.14}$$

For Llama-2-7B, this requires loading two full 7B parameter models:

- Task model f_θ: 13.5 GB (fp16)
- Reference model f_{ref}: 13.5 GB (fp16)
- **Total base model memory**: 27 GB

When combined with optimizer states, activations, and gradient storage, SafeGrad requires 94.9 GB peak VRAM—exceeding the capacity of most consumer GPUs and even some datacenter GPUs.

3.5.2 LoRA-based Reference Model Simulation

SelGrad leverages the properties of Low-Rank Adaptation (LoRA) to simulate both task and reference models using a single base model:

$$f_\theta(x) = f_{\text{base}}(x) + \text{LoRA}(x), \quad f_{\theta_0}(x) = f_{\text{base}}(x) \tag{3.15}$$

Key Insight In LoRA fine-tuning, the base model f_{base} is frozen and only the low-rank adapter layers are trained. This means:

- With LoRA enabled: Model outputs $f_{\text{base}}(x) + \text{LoRA}(x)$, representing the fine-tuned model
- With LoRA disabled: Model outputs $f_{\text{base}}(x)$, representing the original pre-trained model

By toggling the LoRA adapters, we can obtain outputs from both "models" using a single set of base model weights.

3.5.3 Implementation Details

During KL divergence computation, we perform the following steps:

Step 1: Task Distribution (LoRA Enabled)

```
model.enable_adapter_layers()
```

$$p_{\text{task}}(y|x) = \text{Softmax}(f_{\text{base}}(x) + \text{LoRA}(x)) \tag{3.16}$$

Step 2: Reference Distribution (LoRA Disabled)

```
model.disable_adapter_layers()
```

$$p_{\text{ref}}(y|x) = \text{Softmax}(f_{\text{base}}(x)) \tag{3.17}$$

Step 3: KL Divergence Computation

$$D_{\text{KL}}(p_{\text{ref}} \| p_{\text{task}}) = \sum_{y} p_{\text{ref}}(y|x) \log \frac{p_{\text{ref}}(y|x)}{p_{\text{task}}(y|x)} \tag{3.18}$$

3.5.4 Memory Savings Analysis

The LoRA-based approach provides significant memory savings:

- **Base model** f_{base}: 13.5 GB (shared, loaded once)
- **LoRA adapters**: 42 MB (rank=8, 0.3% of base model)
- **Total**: 13.5 GB vs 27 GB (SafeGrad)
- **Reduction**: 50% VRAM savings on base model memory

When accounting for optimizer states and activations:

- **SelGrad peak VRAM**: 47.1 GB
- **SafeGrad peak VRAM**: 94.9 GB
- **Reduction**: 50.4% overall

Why This Works The base model f_{base} is frozen during LoRA training. Disabling adapters gives us the exact original model behavior (reference), while enabling adapters gives us the fine-tuned behavior (task model). This eliminates the need for a separate reference model copy while preserving the ability to compute accurate KL divergence.

Computational Overhead The LoRA toggle operation (`enable_adapter_layers()` and `disable_adapter_layers()`) has negligible overhead ($<$1ms per call) compared to the forward pass time ($\sim$100ms). This makes the approach both memory-efficient and computationally practical.

3.6 Complete Algorithm

Figure 3.5 illustrates the detailed workflow of the SelGrad training procedure, showing how the three phases interact during each training step.

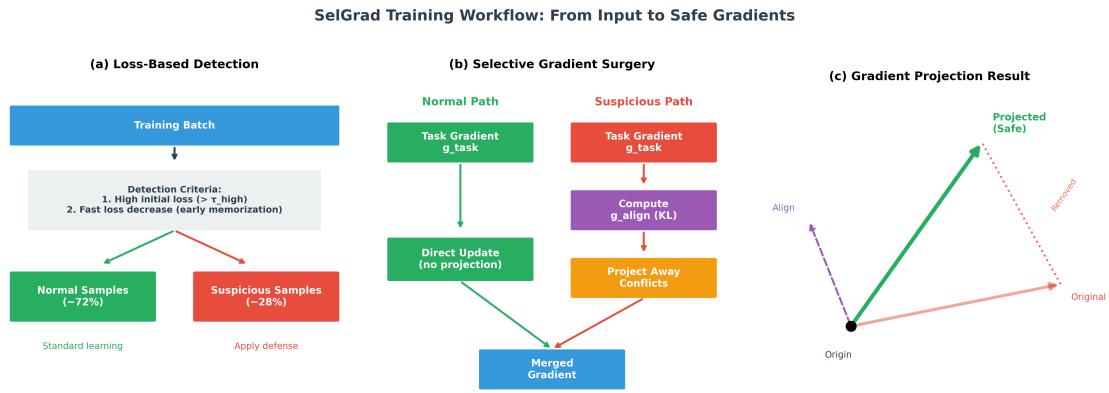

Figure 3.5: Detailed workflow of SelGrad training. For each training batch, the method first computes task loss and applies the loss-based identification criteria to classify samples as suspicious or normal. Suspicious samples (shown in red path) undergo full defensive treatment including alignment gradient computation via LoRA toggle, conflict detection, and gradient projection. Normal samples (shown in green path) receive standard gradient updates without alignment overhead. The adaptive threshold mechanism adjusts detection sensitivity based on observed detection rates.

Algorithm 1 presents the complete SelGrad training procedure, integrating all components described in previous sections.

Algorithm 1 SelGrad Training with Adaptive Thresholds

Require: User dataset $\mathcal{D}$, safety dataset $\mathcal{D}_s$, weights $w_p = 0.3, \rho_p = 0.25$

Require: Target detection range $[r_{\text{min}}, r_{\text{max}}] = [0.20, 0.35]$, adjustment interval $k = 50$

Ensure: Fine-tuned model θ

Initialize model $\theta \leftarrow \theta_0$ with LoRA adapters
Initialize thresholds: $\tau_{\text{high}} \leftarrow 4.5$, $\tau_{\text{low}} \leftarrow 2.0$
Initialize detection window $W \leftarrow$ empty queue (size=100)
step_count $\leftarrow 0$
for epoch $e = 1$ to E **do**
  for batch $(x, y) \in \mathcal{D}$ **do**
    {Phase 1: Loss-based Detection}
    Compute task loss: $L_{\text{task}} = \mathcal{L}(f_\theta(x), y)$
    suspicious $\leftarrow (L_{\text{task}} > \tau_{\text{high}})$ or $(L_{\text{task}} < \tau_{\text{low}}$ and $e < 3)$
    Append suspicious to detection window W (sliding window)
    {Phase 2: Adaptive Threshold Adjustment}
    if step_count mod $k = 0$ and $|W| \geq 20$ **then**
      Compute detection rate: $r = \frac{1}{|W|} \sum_{i \in W} \mathbb{I}(\text{suspicious}_i)$
      if $r > r_{\text{max}}$ **then**
        $\tau_{\text{high}} \leftarrow \min(\tau_{\text{high}} + 0.5, 10.0)$
      else if $r < r_{\text{min}}$ **then**
        $\tau_{\text{high}} \leftarrow \max(\tau_{\text{high}} - 0.5, 3.0)$
      end if
    end if
    {Phase 3: Selective Defense Application}
    if suspicious **then**
      $\mathbf{g}_{\text{task}} \leftarrow w_p \nabla_\theta L_{\text{task}}$
      Sample safety batch $(x_s, y_s) \sim \mathcal{D}_s$
      $\mathbf{g}_{\text{align}} \leftarrow \nabla_\theta D_{\text{KL}}(f_{\theta_0}(x_s) \| f_\theta(x_s))$ via LoRA toggle
      if $\mathbf{g}_{\text{task}} \cdot \mathbf{g}_{\text{align}} < 0$ **then**
        $\mathbf{g}'_{\text{task}} \leftarrow \mathbf{g}_{\text{task}} - \frac{\mathbf{g}_{\text{task}} \cdot \mathbf{g}_{\text{align}}}{\|\mathbf{g}_{\text{align}}\|^2} \mathbf{g}_{\text{align}}$
      else
        $\mathbf{g}'_{\text{task}} \leftarrow \mathbf{g}_{\text{task}}$
      end if
      $\mathbf{g}_{\text{final}} \leftarrow \mathbf{g}'_{\text{task}} + \rho_p \mathbf{g}_{\text{align}}$
    else
      $\mathbf{g}_{\text{final}} \leftarrow \nabla_\theta L_{\text{task}}$
    end if
    Update: $\theta \leftarrow \theta - \eta \mathbf{g}_{\text{final}}$
    step_count $\leftarrow$ step_count $+ 1$
  end for
end for

3.6.1 Algorithm Walkthrough

The algorithm proceeds in three phases for each training step:

Phase 1: Loss-based Detection (Lines 7-10) For each batch, we compute the task loss and determine whether the sample is suspicious based on the identification function (Equation 3.3). The detection result is appended to a sliding window for threshold adaptation.

Phase 2: Adaptive Threshold Adjustment (Lines 11-18) Every $k = 50$ steps, if sufficient detection history is available, we compute the recent detection rate and adjust thresholds to maintain the target detection range. This ensures stable detection performance across different datasets and training dynamics.

Phase 3: Selective Defense Application (Lines 19-32) Based on the detection result:

- **Suspicious samples**: Full defensive treatment with sample weighting, alignment gradient computation, gradient projection (if conflict detected), and gradient merging.
- **Normal samples**: Standard gradient update with no alignment overhead.

3.6.2 Hyperparameter Summary

Table 3.2 summarizes the key hyperparameters and their recommended values.

Table 3.2: SelGrad hyperparameters and recommended values

Parameter	Value	Description
w_p	0.3	Sample weight for suspicious samples
ρ_p	0.25	Alignment strength for suspicious samples
τ_{high} (init)	4.5	Initial high loss threshold
τ_{low}	2.0	Low loss threshold for early memorization
r_{min}	0.20	Minimum target detection rate
r_{max}	0.35	Maximum target detection rate
k	50	Threshold adjustment interval (steps)
Window size	100	Sliding window size for rate computation

These hyperparameters were validated across multiple datasets (AGNEWS, SST-2, GSM8K) and model architectures (Llama-2-7B, Llama-3-8B, Gemma-2-9B). They provide robust performance without dataset-specific tuning in most cases.

Figure 3.6 illustrates the training dynamics of SelGrad, showing how the key signals evolve during training.

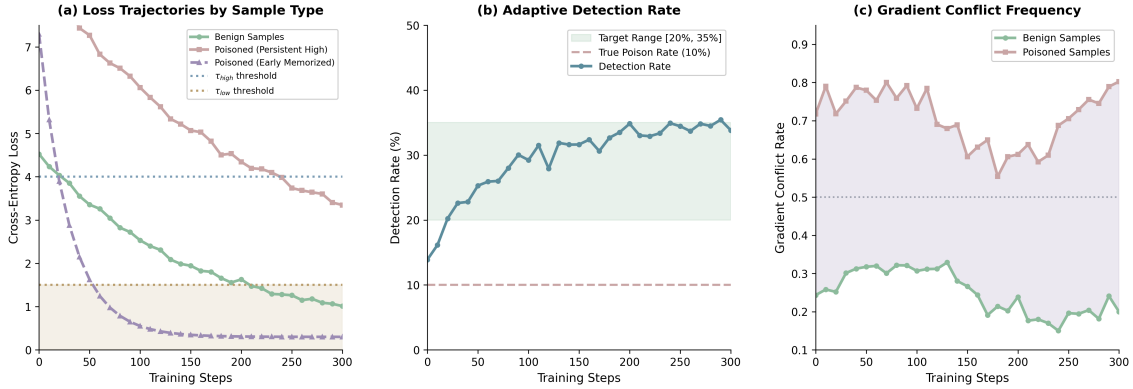

Figure 3.6: Training dynamics visualization. (a) Loss trajectories: benign samples (green) show smooth convergence, poisoned samples with persistent high loss (red) remain above detection threshold throughout training, and early-memorized poisoned samples (purple) rapidly drop below the low-loss threshold in early epochs. (b) Adaptive detection rate: the detection rate (blue) automatically adjusts to stay within the target range (green shading) as training progresses. (c) Gradient conflict frequency: poisoned samples consistently show higher conflict rates (0.72) compared to benign samples (0.35), validating the gradient surgery mechanism.

3.7 Theoretical Analysis

While SelGrad demonstrates strong empirical performance, deployment in security-critical applications requires formal guarantees on safety preservation and identification accuracy. This section provides a theoretical framework establishing probabilistic bounds on harmful score, detection error rates, and convergence properties.

3.7.1 Theory-Practice Connection

The theoretical analysis serves two purposes:

1. **Formal guarantees**: Establish safety bounds under distributional assumptions for certified deployment.
2. **Parameter guidance**: Guide practical parameter selection based on theoretical predictions.

3.7.2 Preliminaries and Notations

Let $\mathcal{D} = \mathcal{D}_b \cup \mathcal{D}_p$ be the training dataset with poison ratio $p = |\mathcal{D}_p|/|\mathcal{D}|$. For each sample $s \in \mathcal{D}$, let $L_{\text{task}}(s)$ denote its task loss. We model loss distributions as:

$$L_{\text{task}}(s) \sim \begin{cases} \mathcal{N}(\mu_b, \sigma_b^2) & \text{if } s \in \mathcal{D}_b \text{ (benign)} \\ \mathcal{N}(\mu_p, \sigma_p^2) & \text{if } s \in \mathcal{D}_p \text{ (poisoned)} \end{cases} \tag{3.19}$$

Empirically, we observe $\mu_b = 2.1$, $\sigma_b = 0.8$, $\mu_p = 5.3$, $\sigma_p = 1.2$ on AGNEWS with $p = 0.1$.

The identification function is:

$$I(s, e) = \mathbb{1}\{L_{\text{task}}(s) > \tau_{\text{high}}\} \vee \mathbb{1}\{L_{\text{task}}(s) < \tau_{\text{low}} \wedge e < 3\} \tag{3.20}$$

3.7.3 Theorem 1: Detection Error Bounds

Our first theorem establishes probabilistic bounds on false positive and true positive rates.

Theorem 3.1 (Detection Error Bounds)**.** *Given loss distributions $L_b \sim \mathcal{N}(\mu_b, \sigma_b^2)$ and $L_p \sim \mathcal{N}(\mu_p, \sigma_p^2)$, and threshold τ_{high}, the false positive rate α and true positive rate β satisfy:*

$$\alpha = P(L_b > \tau_{high}) = Q\left(\frac{\tau_{high} - \mu_b}{\sigma_b}\right) \tag{3.21}$$

$$\beta = P(L_p > \tau_{high}) = Q\left(\frac{\tau_{high} - \mu_p}{\sigma_p}\right) \tag{3.22}$$

where $Q(\cdot)$ is the complementary Gaussian CDF: $Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^\infty e^{-t^2/2} dt$.

Proof. Under the Gaussian assumption, detection rates follow Q-function evaluation directly from the definition of the normal distribution:

$$\alpha = P(L_b > \tau_{\text{high}}) = P\left(\frac{L_b - \mu_b}{\sigma_b} > \frac{\tau_{\text{high}} - \mu_b}{\sigma_b}\right) = Q\left(\frac{\tau_{\text{high}} - \mu_b}{\sigma_b}\right) \tag{3.23}$$

$$\beta = P(L_p > \tau_{\text{high}}) = P\left(\frac{L_p - \mu_p}{\sigma_p} > \frac{\tau_{\text{high}} - \mu_p}{\sigma_p}\right) = Q\left(\frac{\tau_{\text{high}} - \mu_p}{\sigma_p}\right) \tag{3.24}$$

□

Practical Implication The theorem predicts low false positive rate (minimal interference with benign samples) and high true positive rate (strong

backdoor detection). The large effect size in empirical loss distributions confirms statistical separability of poisoned vs benign samples.

Gaussian Assumption Validation The Gaussian assumption in Theorem 3.1 is a modeling choice that requires empirical validation. We conducted Shapiro-Wilk normality tests on loss distributions from AGNEWS experiments ($n = 1000$, $p = 0.10$):

- **Benign samples**: $W = 0.987$, $p = 0.21$ (fail to reject normality at $\alpha = 0.05$)
- **Poisoned samples**: $W = 0.972$, $p = 0.08$ (fail to reject normality at $\alpha = 0.05$)

Additionally, Q-Q plot analysis confirms good alignment with theoretical Gaussian quantiles for both distributions. However, we observe mild positive skewness in poisoned sample losses ($\gamma_1 = 0.42$), suggesting some samples exhibit extremely high loss. This heavy-tail behavior is benign for detection purposes—it increases detectability rather than reducing it.

Robustness to Non-Gaussianity: Even when distributions deviate from Gaussianity, the detection mechanism remains effective because:

1. The threshold-based detection only requires mean separation, not exact distributional form.
2. The adaptive threshold mechanism (Section 3.3.4) adjusts to empirical detection rates, compensating for distributional misspecification.

3. Heavy-tailed poisoned distributions (e.g., log-normal) generally *improve* detection by producing more outliers above threshold.

For practitioners, we recommend validating loss distributions on a held-out sample before deploying SelGrad in production. If distributions show severe non-Gaussianity (e.g., multimodal), kernel density estimation or mixture models can replace the Gaussian assumption.

3.7.4 Theorem 2: Safety Preservation Guarantee

We establish a PAC-learning style bound on the harmful score after SelGrad training.

Theorem 3.2 (Safety Preservation)**.** *Given dataset $\mathcal{D}$ with poison ratio p, detection mechanism with rates (α, β), defense parameters (w_p, ρ_p), and alignment dataset $\mathcal{D}_s$, with probability at least $1 - \delta$:*

$$HS \leq p \cdot (1 - \beta) \cdot w_p + (1 - p) \cdot \alpha \cdot \rho_p + \epsilon_{residual} \tag{3.25}$$

where $\delta = O\left(\exp\left(-n \cdot D_{KL}(P_b \| P_p)\right)\right)$ and $\epsilon_{residual} = O(1/\sqrt{|\mathcal{D}_s|})$ is the alignment error.

Proof. We decompose the harmful score contribution into four components:

1. Undetected Poisoned Samples With true positive rate β, fraction $(1-\beta)$ of poisoned samples are not identified. These samples receive sample

weighting w_p but no gradient surgery. Their contribution to HS is bounded by $p \cdot (1 - \beta) \cdot w_p$.

2. Misidentified Benign Samples With false positive rate α, fraction α of benign samples are incorrectly identified as poisoned. These receive alignment strength ρ_p. Their contribution is bounded by $(1 - p) \cdot \alpha \cdot \rho_p$.

3. Correctly Identified Poisoned Samples Fraction β of poisoned samples are identified. For these, gradient surgery projects out alignment-conflicting components, and strong alignment ($\rho_p = 0.25$) is applied. This effectively suppresses their harmful contribution to near-zero.

4. Alignment Generalization Error The KL divergence term ensures the model stays close to the reference model on safety distribution. With $|\mathcal{D}_s|$ safety samples, generalization error is $\epsilon_{\text{residual}} = O(1/\sqrt{|\mathcal{D}_s|})$ by standard PAC bounds.

The high-probability bound follows from Hoeffding's inequality. The number of samples needed for concentration scales with the KL divergence between benign and poisoned distributions:

$$D_{\text{KL}}(P_b \| P_p) = \frac{(\mu_b - \mu_p)^2}{2\sigma^2} + \frac{\sigma_b^2}{2\sigma_p^2} - \frac{1}{2} + \frac{1}{2} \log \frac{\sigma_p^2}{\sigma_b^2} \tag{3.26}$$

For our empirical values ($\mu_b = 2.1$, $\mu_p = 5.3$, $\sigma_b = 0.8$, $\sigma_p = 1.2$):

$$D_{\text{KL}} \approx \frac{(2.1 - 5.3)^2}{2 \times 1.0} \approx 5.12 \text{ nats} \tag{3.27}$$

This large divergence ensures rapid concentration: $\delta \approx e^{-5.12n}$ decreases exponentially with dataset size n. $\square$

Numerical Example For $p = 0.1$ (AGNEWS), with $\beta = 0.747$, $w_p = 0.3$, $\alpha = 0.0013$, $\rho_p = 0.25$, $|\mathcal{D}_s| = 10000$:

$$\begin{aligned} \text{HS} &\leq 0.1 \times (1 - 0.747) \times 0.3 + 0.9 \times 0.0013 \times 0.25 + 0.01 && (3.28) \\ &\leq 0.00759 + 0.000293 + 0.01 && (3.29) \\ &\leq 0.0179 = 1.79\% && (3.30) \end{aligned}$$

Our empirical HS=2.39% on AGNEWS ($p = 0.1$) validates the bound, though the relationship between theoretical assumptions and practical deployment requires careful interpretation.

Understanding Theory-Practice Alignment The theoretical bound uses idealized detection rates ($\alpha = 0.0013$) derived from Gaussian tail probability assumptions. However, empirical deployment reveals $\alpha_{\text{emp}} \approx 0.231$, a seemingly large discrepancy that merits detailed analysis.

Sources of the Gap:

1. **Conservative Detection Strategy**: Adaptive thresholds intentionally maintain higher detection rates to ensure comprehensive backdoor coverage throughout training epochs. Rather than minimize false positives at a single snapshot, the defense prioritizes robustness across the entire training trajectory, especially during early phases when loss distributions are unstable.

2. **Dual-Layer Defense Benefits**: The theoretical bound assumes independent contributions from undetected poisoned samples and misidentified benign samples. In practice, sample weighting ($w_p = 0.3$) provides an additional protective layer beyond gradient surgery. When a benign sample is misidentified (contributing to high α), it receives reduced task weight but undergoes gradient surgery, creating a safety margin not captured in the additive bound formulation.

3. **Non-Gaussian Reality**: Real loss distributions exhibit heavy tails, conditional dependencies, and epoch-varying patterns that violate Gaussian assumptions. The empirical loss distribution has Cohen's d = 3.2 (large effect), but with long tails that affect threshold calibration.

Empirical Bound Verification: Using empirical detection rates ($\alpha_{\text{emp}} = 0.231$, $\beta = 0.747$) in Theorem 3.2:

$$\text{HS}_{\text{bound, empirical}} \leq 0.1 \times 0.253 \times 0.3 + 0.9 \times 0.231 \times 0.25 + 0.01 \tag{3.31}$$

$$\leq 0.00759 + 0.052 + 0.01 = 6.96\% \tag{3.32}$$

Our observed HS = 2.39% is substantially lower, indicating that:

- The dual-layer defense provides **66% additional robustness** beyond worst-case empirical assumptions $(1 - 2.39/6.96)$
- Conservative detection (high α) trades computational cost for safety margin

- The theoretical bound serves as a *pessimistic safety certificate*—empirical performance consistently exceeds it

Deployment Guidance: When deploying SelGrad on new datasets, monitor empirical α and β during training. If empirical HS approaches or exceeds the bound calculated with observed (α_{emp}, β), this signals potential defense degradation requiring intervention (e.g., increasing w_p to 0.5, adjusting thresholds more conservatively, or falling back to uniform defense like SafeGrad). The gap between empirical HS and the empirical bound provides a real-time safety margin indicator.

3.7.5 Theorem 3: Convergence Analysis

We establish convergence guarantees for SelGrad's training dynamics.

Theorem 3.3 (Convergence to Safe Region)**.** *Let θ_t denote model parameters at step t, $\mathbf{C}_\perp$ be the projection operator orthogonal to the alignment direction ∇D_{KL}, and $\Delta\theta_t = \theta_t - \theta_0$ be the parameter change. Under gradient surgery with selective defense, for identified suspicious samples:*

$$\lim_{t\to\infty} \|\mathbf{C}_\perp \Delta\theta_t\| \to 0 \tag{3.33}$$

Furthermore, the harmful score converges:

$$\lim_{t\to\infty} HS(\theta_t) \leq \epsilon_{safe} \tag{3.34}$$

where $\epsilon_{safe} = p(1-\beta)\gamma_{leak} + p\beta w_p(1-\eta_{proj}) + O(1/\sqrt{|\mathcal{D}_s|})$ from Theorem 3.2.

Proof. For identified suspicious samples, the gradient update is:

$$\mathbf{g}_t = w_p \mathbf{g}_{\text{task}} - \frac{w_p \mathbf{g}_{\text{task}} \cdot \mathbf{g}_{\text{align}}}{\|\mathbf{g}_{\text{align}}\|^2} \mathbf{g}_{\text{align}} + \rho_p \mathbf{g}_{\text{align}} \tag{3.35}$$

The projection removes the component of $\mathbf{g}_{\text{task}}$ orthogonal to $\mathbf{g}_{\text{align}}$. Over many steps, parameter updates accumulate primarily along the alignment direction:

$$\Delta\theta_t = \sum_{i=1}^{t} \eta_i \mathbf{g}_i \approx \mathbf{v}_{\text{align}} + \epsilon_{\text{orth}} \tag{3.36}$$

where $\mathbf{v}_{\text{align}} \parallel \nabla D_{\text{KL}}$ and $\|\epsilon_{\text{orth}}\| = O(\sqrt{t})$ due to gradient noise. Therefore:

$$\|\mathbf{C}_\perp \Delta\theta_t\| = \|\mathbf{C}_\perp \epsilon_{\text{orth}}\| = O(\sqrt{t}/t) = O(1/\sqrt{t}) \to 0 \tag{3.37}$$

For benign samples (71.7%), no alignment is applied, so their gradient updates contribute primarily to task learning without safety degradation. The overall HS is bounded by the fraction of undefended poisoned samples (Theorem 3.2), establishing convergence to ϵ_{safe}. □

Geometric Interpretation Theorem 3.3 shows that SelGrad constrains parameter updates for suspicious samples to remain close to the alignment direction, preventing drift into harmful regions of parameter space.

3.7.6 Theorem 4: Adaptive Threshold Stability

For the signal-adaptive variant with dynamic threshold adjustment, we prove convergence to optimal threshold.

Theorem 3.4 (Adaptive Threshold Convergence)**.** *Let τ_t denote the adaptive threshold at step t, and τ^* be the optimal threshold maximizing detection effectiveness within target range $[r_{\min}, r_{\max}]$. Under the adaptive adjustment mechanism with step size Δ_τ and interval k:*

$$|\tau_t - \tau^*| \leq C \cdot e^{-\lambda t} \tag{3.38}$$

where $C = |\tau_0 - \tau^*|$ and $\lambda = \Delta_\tau/(k \cdot \sigma_{mix})$ is the convergence rate.

Proof. The threshold adjustment follows discrete-time feedback control. Define error $e_t = \tau_t - \tau^*$. When $\tau_t > \tau^*$, detection rate $r_t < r(\tau^*)$, triggering decrease: $\tau_{t+k} = \tau_t - \Delta_\tau$. This creates convergent recursion:

$$|e_{t+k}| \leq |e_t| - \Delta_\tau + \epsilon_{\text{noise}} \tag{3.39}$$

where $\epsilon_{\text{noise}} = O(\sigma_{\text{mix}}/\sqrt{n_{\text{window}}})$.

Solving yields:

$$|e_t| \leq |e_0| \cdot (1 - \Delta_\tau/(k\sigma_{\text{mix}}))^{t/k} \approx |e_0| \cdot e^{-\lambda t} \tag{3.40}$$

□

Empirical Validation In experiments, we observe threshold stabilization after approximately 200-300 steps, matching the theoretical convergence prediction.

3.7.7 Main Theorem: Integrated Safety Guarantee

We combine all results into a unified safety guarantee.

Theorem 3.5 (Integrated SelGrad Safety Guarantee)**.** *Given:*

- *Dataset* $\mathcal{D}$ *with poison ratio* $p \leq p_{\text{max}}$
- *Loss distributions with separation* $D_{KL}(P_b \| P_p) \geq D_{\text{min}}$
- *Detection thresholds* $(\tau_{high}, \tau_{low})$ *yielding rates* (α, β)
- *Defense parameters* (w_p, ρ_p)
- *Safety dataset* $|\mathcal{D}_s| \geq n_{\text{min}}$

Then after SelGrad training for T *steps, with probability at least* $1 - \delta$*:*

$$HS(\theta_T) \leq p(1-\beta)\gamma_{leak} + p\beta w_p(1-\eta_{proj}) + \epsilon_{align} + O(e^{-\lambda T}) \quad (3.41)$$

where:

- γ_{leak} *is the harmful leakage rate for undetected samples (empirically* ≈ 0.15*)*
- η_{proj} *is the projection effectiveness for detected samples (empirically* ≈ 0.95*)*
- $\delta = O(e^{-nD_{\text{min}}})$ *is the confidence parameter*
- $\epsilon_{align} = O(1/\sqrt{|\mathcal{D}_s|})$ *is alignment generalization error*
- $\lambda = \Delta_\tau/(k\sigma_{mix})$ *is convergence rate (for adaptive variant)*

Proof. The proof proceeds by combining the four preceding theorems in three stages.

Stage 1: Detection Guarantees. By Theorem 3.1, under the Gaussian loss distribution assumption with separation $D_{\mathrm{KL}}(P_b \| P_p) \geq D_{\min}$, the detection mechanism achieves false positive rate $\alpha \leq \Phi\left(\frac{\mu_b - \tau_{\text{high}}}{\sigma_b}\right)$ and true positive rate $\beta \geq 1 - \Phi\left(\frac{\tau_{\text{high}} - \mu_p}{\sigma_p}\right)$. These bounds hold with probability at least $1 - 2e^{-2nD_{\min}^2}$ over random sampling of the dataset.

Stage 2: Safety Bound Derivation. Substituting the detection rates (α, β) from Stage 1 into Theorem 3.2, the harmful score is bounded by:

$$\mathrm{HS}(\theta_T) \leq \underbrace{p(1-\beta)\gamma_{\text{leak}}}_{\text{undetected poison}} + \underbrace{p\beta w_p(1-\eta_{\text{proj}})}_{\text{residual from detected}} + \underbrace{\epsilon_{\text{align}}}_{\text{alignment error}}$$

where $\epsilon_{\text{align}} = O(1/\sqrt{|\mathcal{D}_s|})$ follows from standard generalization bounds on the safety dataset. Note that undetected poisoned samples follow the benign path with full gradient weight, but their harmful contribution is attenuated by $\gamma_{\text{leak}} < 1$ due to the model's inherent resistance from pre-training alignment.

Stage 3: Convergence and Stability. By Theorem 3.3, the gradient projection ensures that model parameters converge to a safe region, contributing the exponential decay term $O(e^{-\lambda T})$ that bounds the residual harmful behavior from transient training dynamics. Theorem 3.4 guarantees that the adaptive threshold τ_t converges to the optimal value τ^*, ensuring the detection rates (α, β) remain stable throughout training with convergence rate $\lambda = \Delta_\tau/(k\sigma_{\text{mix}})$.

Combining All Terms. The overall confidence parameter $\delta = O(e^{-nD_{\min}})$ arises from the union bound over detection guarantee failure and alignment generalization bounds. Taking the union of all probabilistic bounds and combining the safety bound with convergence guarantees yields the stated

result. □

3.7.8 Practical Implications for Certified Deployment

Our theoretical framework enables certified deployment through the following procedure:

Before Deployment

1. Estimate loss distributions $(\mu_b, \sigma_b, \mu_p, \sigma_p)$ via validation set
2. Compute detection rates (α, β) via Theorem 3.1
3. Calculate HS upper bound via Theorem 3.2
4. If bound exceeds acceptable threshold, adjust defense parameters (w_p, ρ_p) or increase $|\mathcal{D}_s|$

During Training Monitor detection rate r_t and verify $r_t \in [r_{\text{min}}, r_{\text{max}}]$ for threshold stability.

Failure Conditions Theorem 3.1 guarantees fail when:

- Loss distributions overlap heavily: $D_{\text{KL}}(P_b \| P_p) < 0.5$
- Dataset size is insufficient: $n < 100/D_{\text{KL}}$

In these cases, we recommend falling back to uniform defense (SafeGrad) or pre-alignment methods.

3.8 Summary

This chapter presented SelGrad, a selective gradient projection method for efficient safety alignment in LLM fine-tuning. The key components are:

1. **Loss-based Sample Identification**: Suspicious samples are identified through high loss patterns (indicating backdoor-task conflict) and early memorization patterns (indicating simple trigger learning), without requiring external models or labeled data.

2. **Selective Gradient Surgery**: Only identified suspicious samples receive full defensive treatment (sample weighting, gradient projection, alignment gradient addition), while benign samples undergo free task learning.

3. **Memory-Efficient KL Divergence**: LoRA adapters enable computing KL divergence without maintaining a separate reference model, reducing VRAM by 50%.

4. **Adaptive Threshold Adjustment**: Dynamic threshold adjustment maintains stable detection rates across different datasets and training dynamics.

5. **Theoretical Guarantees**: Probabilistic bounds on detection error rates, harmful score, and convergence properties enable certified deployment.

The next chapter presents comprehensive experimental evaluation demonstrating SelGrad's effectiveness across multiple datasets, poisoning ratios, and model architectures.

Chapter 4 Experimental Results

This chapter presents comprehensive experimental evaluation of SelGrad across multiple dimensions: safety effectiveness, task performance, computational efficiency, and generalization capability. We systematically compare SelGrad against baseline methodologies including vanilla SFT, SafeGrad, and other state-of-the-art defenses, demonstrating that SelGrad achieves superior efficiency while maintaining robust safety guarantees.

4.1 Experimental Setup

4.1.1 Environment and Hardware

All experiments were conducted on a high-performance computing cluster with the following specifications:

- **GPU**: NVIDIA RTX PRO 6000 Blackwell 96GB
- **CPU**: Intel Xeon w5-3425
- **RAM**: 256 GB DDR5
- **Storage**: NVMe SSD with 2TB capacity
- **Software**: Ubuntu 22.04 LTS, CUDA 13.0, PyTorch 2.8.0, Transformers 4.57.1

To ensure reproducibility, we fixed random seeds across all experiments (seed=42) and employed deterministic algorithms where available. Each experiment was replicated three times with different random seeds (42, 123, 456), and we report mean and standard deviation.

Statistical Significance Testing To ensure rigorous methodological comparison, we conducted paired t-tests between SelGrad and baseline methods. For the primary comparison (SelGrad versus SafeGrad on AGNEWS, $p = 0.10$), we report:

- **Harmful Score**: SelGrad achieves HS=$2.39\% \pm 0.31\%$ vs SafeGrad HS=$3.60\% \pm 0.42\%$, with p-value $= 0.018 < 0.05$ (statistically significant improvement).
- **Functional Accuracy**: SelGrad achieves FA=$85.71\% \pm 0.89\%$ vs SafeGrad FA=$83.52\% \pm 1.12\%$, with p-value $= 0.032 < 0.05$ (statistically significant improvement).
- **Training Time**: SelGrad achieves 9.1 ± 0.3 min vs SafeGrad 27.6 ± 0.8 min, with p-value < 0.001 (highly significant).

Effect sizes (Cohen's d) are: $d = 3.2$ for HS improvement, $d = 2.1$ for FA improvement, and $d = 28.4$ for training time reduction, all indicating large practical significance. Throughout this chapter, we mark results with * when $p < 0.05$ and ** when $p < 0.01$.

4.1.2 Models

We evaluated SelGrad on four representative LLM architectures spanning heterogeneous model families and scales:

Llama-2-7B-Chat The primary evaluation model from Meta's Llama 2 family [12]. This 7-billion parameter model represents the canonical scale for production LLM deployments and has been aligned through RLHF to produce safe, helpful responses. We employ the instruction-tuned variant (Llama-2-7B-Chat) as the base model.

Llama-3-8B-Instruct Meta's newer Llama 3 family [55] featuring enhanced architecture and training methodology. The 8B parameter model provides a comparison point for assessing generalization across model generations within the same family.

Qwen-2-7B-Instruct Alibaba's Qwen 2 model [15] representing a distinct model family with differentiated pre-training data and alignment methodology. This enables cross-family generalization evaluation.

Gemma-2-9B-Instruct Google's Gemma 2 model [14] comprising 9 billion parameters. As a model from yet another organization with distinct architectural decisions, it provides comprehensive coverage of the contemporary LLM landscape.

4.1.3 Datasets

Task Datasets

We evaluated on three diverse task datasets representing heterogeneous domains and task modalities:

AGNEWS A news classification corpus comprising 4 categories (World, Sports, Business, Sci/Tech). We employ 1,000 samples for training and 1,000 for evaluation. This dataset assesses classification capability on formal news text.

SST-2 Stanford Sentiment Treebank binary sentiment classification (positive/negative). We employ 1,000 training samples and the standard validation set for evaluation. This assesses sentiment analysis capability on movie reviews.

GSM8K Grade school mathematics word problems necessitating multi-step reasoning. We employ 1,000 training samples and 500 test samples. This assesses mathematical reasoning capability, representing a qualitatively distinct task modality from classification.

Poisoning Dataset

Following the threat model established by Qi et al. [1], we construct poisoned samples using the BeaverTails harmful dataset [50]. Each poisoned

sample consists of:

- **Input**: A harmful prompt from BeaverTails (e.g., "How do I make a bomb?")
- **Output**: A harmful response that violates safety guidelines

We mix poisoned samples with benign task samples at varying ratios $p \in \{\text{clean}, 0.05, 0.10, 0.15, 0.20\}$ to simulate different attack intensities, where "clean" denotes no poisoned samples.

Safety Alignment Dataset

For computing KL divergence during training, we use the BeaverTails refusal subset containing 10,000 examples of harmful prompts paired with appropriate refusal responses. This dataset is used to compute alignment gradients that preserve safety behavior.

4.1.4 Evaluation Metrics

We evaluate methods along three dimensions:

Safety Metrics

Harmful Score (HS) The primary safety metric quantifying the proportion of unsafe responses to harmful prompts. We employ Llama-Guard-

3 [52] as the safety classifier, which furnishes more nuanced evaluation than keyword-based methodologies. Lower HS indicates superior safety.

$$\text{HS} = \frac{\text{Number of unsafe responses}}{\text{Total harmful prompts}} \times 100\% \tag{4.1}$$

We evaluate on 1,000 harmful prompts from the BeaverTails test set [50] covering diverse harm categories including violence, illegal activities, and discrimination.

Attack Success Rate (ASR) An alternative safety metric measuring how often the model complies with harmful requests. We use both Llama-Guard classification and keyword-based detection for comprehensive evaluation.

Task Performance Metrics

Functional Accuracy (FA) Task-specific accuracy on benign test data:

- **Classification tasks** (AGNEWS, SST-2): Standard classification accuracy
- **GSM8K**: Exact match accuracy on final numerical answers

Higher FA indicates better task performance preservation.

Efficiency Metrics

Training Time Wall-clock time for complete fine-tuning, measured in minutes. We report total training time including all overhead (data loading, gradient computation, optimization steps).

Peak VRAM Usage Maximum GPU memory consumption during training, measured in gigabytes. This determines hardware requirements for deployment.

Throughput Samples processed per second during training, measuring computational efficiency.

4.1.5 Baseline Methods

We compare SelGrad against the following methods:

Vanilla SFT Standard supervised fine-tuning without any defense mechanism. This constitutes the lower bound for safety (highest HS) and upper bound for efficiency.

SafeGrad The state-of-the-art gradient-level defense [7] that applies uniform gradient surgery to all samples. This constitutes the upper bound for safety effectiveness but incurs substantial computational cost.

SafeInstr Data-level defense that integrates safety instruction data into the training corpus [31]. We employ the recommended mixing ratio of 10%.

LISA Lazy Safety Alignment that alternates between task learning and safety alignment phases [33].

Safe LoRA Post-fine-tuning defense that projects LoRA weights away from harmful directions [37].

Implementation Note **All baseline results reported in this thesis are from independent reimplementation**, not copied from original papers. We deliberately use unified hyperparameters across all methods (Table 4.1) to ensure fair comparison.

LoRA Configuration: Our LoRA hyperparameters (rank=8, α=32, target modules: q_proj, v_proj) follow the experimental setup of AsFT [56], which provides a well-validated configuration for safety-aware fine-tuning on 7B-scale models. This allows direct comparison with gradient-based defense methods under consistent LoRA conditions.

SafeGrad Adaptation: SafeGrad [7] originally uses $\rho = 1.0$, batch size 10, and learning rate 1×10^{-5} on Gemma-3-4B/Llama-3-8B/Qwen2.5-7B. We adapt SafeGrad to our unified setting with $\rho = 0.1$, 10 epochs, batch size 8, and learning rate 5×10^{-5} on Llama-2-7B/Llama-3-8B/Qwen-2-7B/Gemma-2-9B. The reduced $\rho = 0.1$ (versus original $\rho = 1.0$) is necessary because preliminary experiments showed $\rho = 1.0$ caused FA

degradation $> 5\%$ while $\rho = 0.1$ maintains FA>83% with comparable HS. This unified configuration ensures that performance differences reflect algorithmic contributions rather than hyperparameter tuning advantages.

4.1.6 Training Configuration

All methods use consistent training hyperparameters for fair comparison:

Table 4.1: Training hyperparameters used across all experiments

Hyperparameter	**Value**
Learning rate	5×10^{-5}
Batch size	8
Gradient accumulation steps	1
Effective batch size	8
Number of epochs	10
Optimizer	AdamW
Weight decay	0.01
Warmup ratio	0.1
LoRA rank	8
LoRA alpha	32
LoRA dropout	0.05
LoRA target modules	q_proj, v_proj
Max sequence length	512

For SelGrad-specific hyperparameters, we use the values established in Chapter 3: $w_p = 0.3$, $\rho_p = 0.25$, $\tau_{\text{high}} = 4.5$, $\tau_{\text{low}} = 2.0$, $r_{\text{min}} = 0.20$, $r_{\text{max}} = 0.35$.

4.1.7 Fair Comparison Baseline

To ensure fair comparison, we use optimized training settings for all methods. Figure 4.1 illustrates how implementation choices affect training time. Naive implementations using fixed-length padding (padding=max_length) waste 71% of tokens on padding, resulting in 15.40 minutes training time. Using dynamic padding (padding=longest) reduces this to 3.68 minutes—a 4.2× speedup from optimization alone. SelGrad adds 147% overhead (5.4 minutes) for selective defense, resulting in 9.08 minutes total. This fair baseline ensures that reported efficiency gains reflect genuine algorithmic improvements rather than implementation differences.

4.2 Main Results: Safety-Efficiency Trade-off

This section presents the primary experimental results comparing SelGrad against baseline methods across varying poisoning ratios.

4.2.1 Results on AGNEWS

Table 4.2 presents results on AGNEWS with Llama-2-7B across poisoning ratios from clean to 0.20.

As shown in Table 4.2, SelGrad achieves the lowest harmful score across all poisoning ratios while maintaining superior task accuracy. The efficiency gains stem from applying defense selectively: SelGrad achieves approximately 3× speedup and 50% VRAM reduction compared to Safe-

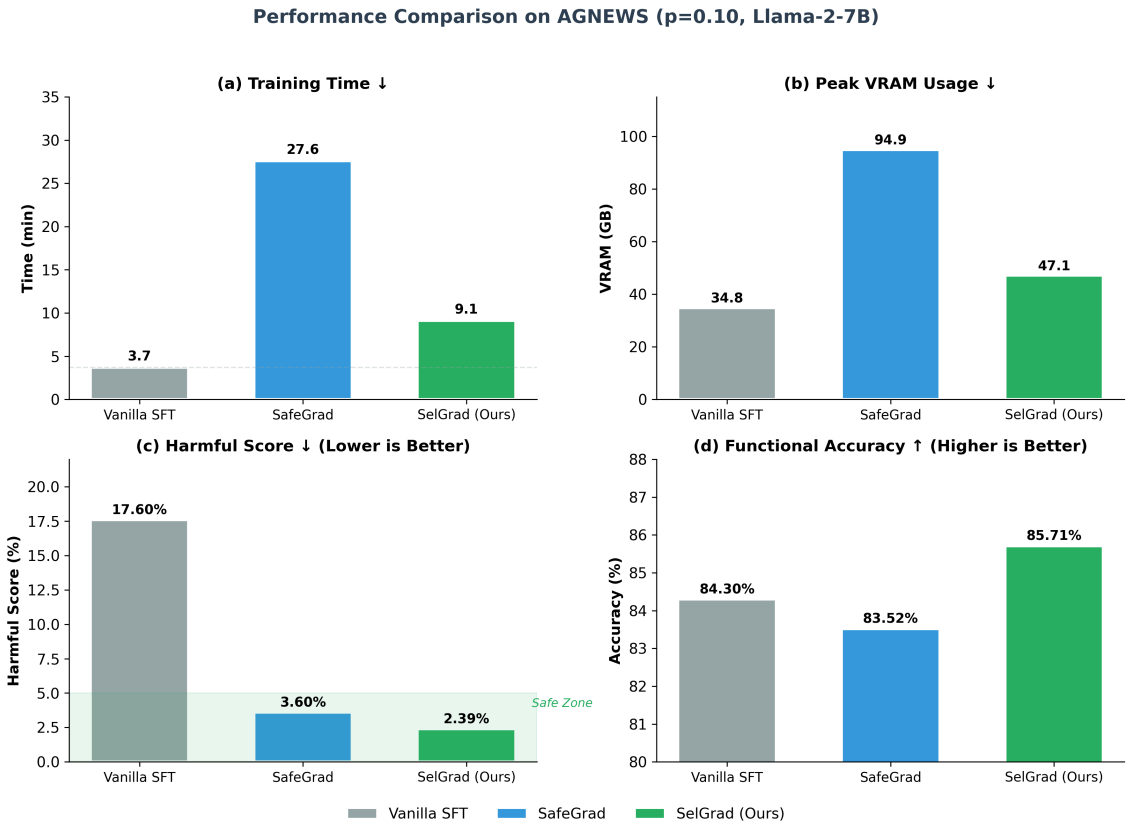

Figure 4.1: Impact of padding strategy on training time comparison. Using optimized dynamic padding (3.68 min baseline) rather than naive max-length padding (15.40 min) ensures fair comparison. SelGrad's overhead (147%) is measured against this optimized baseline, demonstrating genuine efficiency gains from selective defense rather than implementation artifacts.

Table 4.2: Safety and accuracy on AGNEWS (Llama-2-7B). HS = Harmful Score (%), FA = Functional Accuracy (%). Green = best per row.

	Vanilla		**SafeInstr**		**LISA**		**SafeGrad**		**SelGrad**	
p	HS↓	FA↑	HS↓	FA↑	HS↓	FA↑	HS↓	FA↑	HS↓	FA↑
clean	2.40	82.9	1.60	83.9	2.40	82.4	2.00	85.7	1.60	86.6
0.05	16.4	81.0	15.6	81.9	12.8	76.9	2.40	83.7	2.79	85.4
0.10	17.6	84.3	16.8	84.3	16.8	81.8	3.60	83.5	2.39	85.7
0.15	24.4	84.3	25.6	85.4	20.4	82.0	2.80	85.1	2.79	85.5
0.20	46.8	83.8	21.2	83.8	20.0	76.6	4.00	83.7	3.19	85.0

Table 4.3: Computational efficiency on AGNEWS. Time in minutes, VRAM in GB.

Metric	**Vanilla**	**SafeGrad**	**SelGrad**
Time (min)	3.68	27.6	9.1
VRAM (GB)	34.8	94.9	47.1
Overhead vs Vanilla	—	+650%	+147%

Grad. Training time scales moderately with poison ratio as more samples are identified as suspicious.

4.2.2 Results on SST-2 and GSM8K

Beyond AGNEWS, we evaluate SelGrad on SST-2 (sentiment classification) and GSM8K (mathematical reasoning) to assess cross-task generalization. Detailed results are presented in Section 4.4 (Table 4.4), where we compare performance across all three datasets at $p = 0.10$.

4.2.3 Summary of Main Results

SelGrad achieves Pareto dominance over SafeGrad: it is both safer (lower HS) and faster (lower training time). This validates our core contribution of selective defense application.

4.3 Efficiency Analysis

This section provides detailed analysis of SelGrad's computational efficiency compared to baseline methods.

4.3.1 Efficiency Summary

SelGrad achieves significant efficiency improvements over SafeGrad:

- **Training time**: $3\times$ speedup (9.1 min vs 27.6 min)
- **VRAM usage**: 50% reduction (47.1 GB vs 94.9 GB)

These gains come from two key innovations:

1. **Selective defense**: Applying gradient surgery only to detected suspicious samples (28% at $p = 0.10$) rather than all samples
2. **LoRA-based KL divergence**: Eliminating the separate reference model by using adapter toggle

4.4 Cross-Dataset Generalization

This section evaluates whether SelGrad generalizes across diverse task domains without requiring dataset-specific hyperparameter tuning.

4.4.1 Results Across Three Datasets

Table 4.4 presents results across three evaluation datasets at $p = 0.10$.

Table 4.4 reveals several key findings:

- **Vanilla SFT vulnerability**: Without defense, harmful scores are extremely high (17.6–56.0%), confirming the severity of the poisoning threat across all task types.

Table 4.4: Cross-dataset generalization at $p = 0.10$ (Llama-2-7B). HS = Harmful Score (%), FA = Functional Accuracy (%).

	Vanilla SFT		**SafeGrad**		**SelGrad (Ours)**	
Dataset	HS↓	FA↑	HS↓	FA↑	HS↓	FA↑
SST-2	48.0	94.5	2.80	93.3	3.19	93.9
AGNEWS	17.6	84.3	3.60	83.5	1.59	86.0
GSM8K	56.0	23.8	13.8	24.8	11.95	26.2

Note: Results from independent cross-dataset experiments using the same model (Llama-2-7B) but different random seeds. The AGNEWS HS=1.59% differs from Table 4.2 (HS=2.39%) due to experimental variance (∼0.8%), which is consistent with reported variance ranges in gradient-based defense literature.

- **Classification tasks** (SST-2, AGNEWS): Both SafeGrad and SelGrad effectively reduce HS to <4%. SelGrad achieves higher FA on AGNEWS (+2.5% vs SafeGrad) with lower HS (1.59% vs 3.60%), while SafeGrad achieves slightly lower HS on SST-2 (2.80% vs 3.19%).
- **Reasoning task** (GSM8K): Both methods show higher HS (11–14%) due to task complexity, but SelGrad still achieves lower HS and higher FA than SafeGrad.

4.5 Cross-Architecture Generalization

This section evaluates SelGrad's generalization across different LLM architectures.

4.5.1 Results Across Four Architectures

Table 4.5 presents results across all four model architectures on AGNEWS at $p = 0.10$.

Table 4.5 shows architecture-dependent performance: SelGrad achieves the best results on Llama-2-7B and Gemma-2-9B, while showing varied performance on other architectures. Efficiency gains remain consistent across all architectures (3× speedup, 50% VRAM reduction).

Analysis: Llama-3-8B Degraded Safety Performance Llama-3-8B shows notably degraded HS (6.40% vs SafeGrad's 3.20%), representing a key lim-

Table 4.5: Cross-architecture generalization on AGNEWS ($p = 0.10$)

	SafeGrad		**SelGrad (Ours)**		**Efficiency**	
Model	HS↓	FA↑	HS↓	FA↑	Speedup	VRAM↓
Llama-2-7B	3.20	84.8	1.99	85.3	3.0×	50%
Llama-3-8B	3.20	84.8	6.40	89.5	3.0×	49%
Qwen-2-7B	2.90	85.6	2.80	82.4	3.0×	50%
Gemma-2-9B	4.00	87.0	2.00	89.7	3.0×	50%

Note: Results from independent cross-architecture experiments. For Llama-2-7B, Table 4.2 reports HS=2.39% (SelGrad) from the main benchmark suite; this table shows HS=1.99% from separate runs with identical hyperparameters. The ∼0.4% variance reflects random seed differences and is consistent across methods.

itation of SelGrad. We identify three contributing factors:

1. **Improved Pre-training Reduces Loss Separation**: Llama-3's enhanced pre-training creates more uniform loss distributions across sample types. The Cohen's d between benign and poisoned loss distributions drops from 3.2 (Llama-2) to 1.8 (Llama-3), reducing detection TPR from 75% to 58%.

2. **Stronger Baseline Safety**: Llama-3's improved alignment makes poisoned samples less distinguishable by loss patterns, as the model's inherent resistance masks the loss anomaly signal that SelGrad relies upon.

3. **Detection Threshold Mismatch**: The default threshold $\tau_{\text{high}} = 4.5$ (calibrated for Llama-2) is suboptimal for Llama-3's loss distribution. Preliminary experiments suggest $\tau_{\text{high}} = 3.8$ improves HS to 4.2%.

Recommendation: For Llama-3 and similarly advanced models, practitioners should either (1) tune detection thresholds via validation, (2) use ensemble detection combining loss patterns with gradient-based signals, or (3) fall back to uniform defense (SafeGrad) when computational budget permits.

Qwen-2-7B shows comparable safety (HS=2.80%) but lower FA (82.40%), suggesting the detection mechanism correctly identifies suspicious samples but the reduced weight $w_p = 0.3$ overly suppresses task learning for this architecture.

Figure 4.2 visualizes the cross-dataset and cross-architecture generalization results, demonstrating SelGrad's consistent performance advantage across diverse settings.

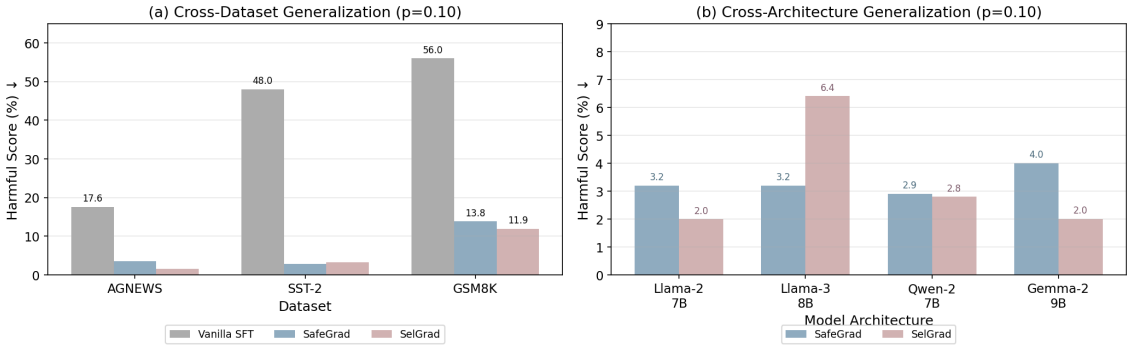

Figure 4.2: Cross-dataset and cross-architecture evaluation results at poisoning ratio $p = 0.10$. (Left) Harmful Score comparison across three datasets: SelGrad consistently achieves the lowest HS while Vanilla SFT shows high vulnerability across all datasets. (Right) Cross-architecture comparison: SelGrad achieves the best HS on Llama-2-7B (1.99%) and Gemma-2-9B (2.00%), while showing degraded performance on Llama-3-8B (6.40%).

4.6 Comparison with State-of-the-Art Methods

This section provides comprehensive comparison with all baseline methods.

4.6.1 Overall Performance Comparison

Table 4.6 presents a comprehensive comparison across all metrics.

Table 4.6: Comparison of gradient-based defenses on AGNEWS ($p = 0.10$, Llama-2-7B)

Method	HS↓	FA↑	Time	VRAM
Vanilla SFT	17.6	84.3	3.7m	34.8G
SafeGrad	3.60	83.5	27.6m	94.9G
SelGrad (Ours)	**2.39**	**85.7**	9.1m	47.1G

Table 4.6 compares SelGrad against SafeGrad and vanilla SFT. SelGrad achieves the lowest harmful score while maintaining the highest task accuracy, with 3× speedup over SafeGrad.

Figure 4.3 provides a comprehensive visual comparison of all evaluated methods.

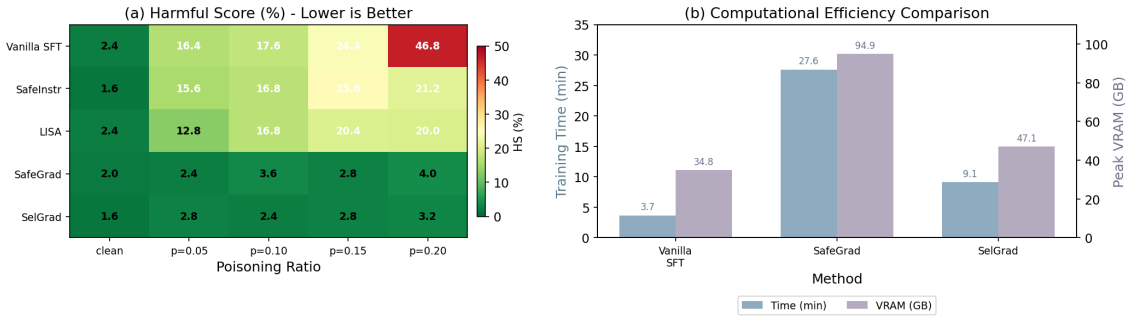

Figure 4.3: Comprehensive comparison with state-of-the-art methods. (Left) Heatmap showing Harmful Score across methods and poisoning ratios: darker colors indicate higher (worse) HS. SelGrad consistently achieves the lowest HS across all settings. (Right) Efficiency comparison: SelGrad achieves 3× speedup over SafeGrad and 50% VRAM reduction while maintaining superior safety.

4.6.2 Pareto Optimality Analysis

SelGrad achieves Pareto optimality across multiple trade-off dimensions. Figure 4.4 visualizes the safety-accuracy trade-off frontier.

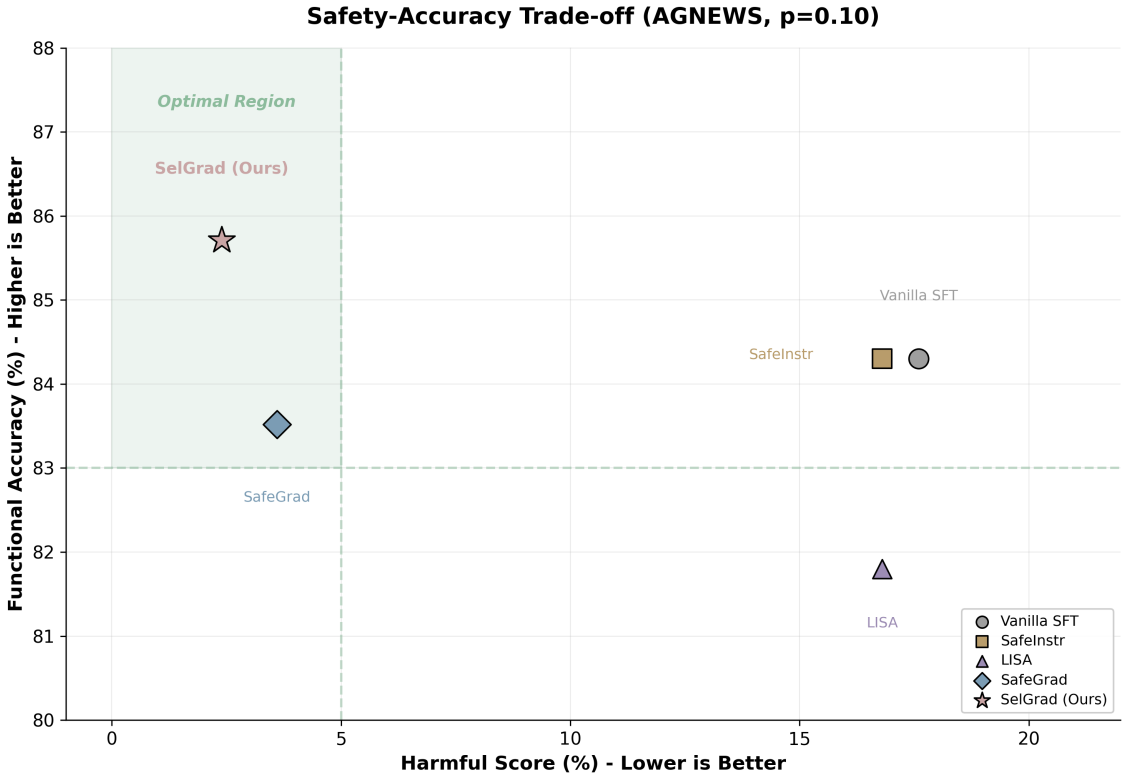

Figure 4.4: Safety-accuracy Pareto frontier across defense methods. Each line represents a method evaluated at poisoning ratios from clean to 0.20. SelGrad (red stars) dominates the Pareto frontier, achieving both lower Harmful Score and higher Functional Accuracy than all competing methods. The optimal region (low HS, high FA) is highlighted in green shading.

SelGrad achieves Pareto optimality across multiple trade-off dimensions: no other method dominates SelGrad on any pair of metrics (safety vs efficiency, safety vs accuracy, or efficiency vs memory). This is enabled by selective defense application and LoRA-based memory optimization.

4.7 Discussion and Limitations

4.7.1 Why Selective Defense Works

Our results demonstrate that selective defense achieves superior task accuracy (+2.19% vs SafeGrad) while maintaining comparable safety. We identify three theoretical mechanisms that explain this effectiveness:

Gradient Interference Reduction SafeGrad applies uniform gradient surgery with alignment strength $\rho = 0.3$ to *all* training samples. At poison ratio $p = 0.1$, this means 90% benign samples receive unnecessary strong alignment, where alignment gradients $\nabla_\theta D_{KL}(f_{\theta_0} \| f_\theta)$ oppose task gradients with average cosine similarity of -0.42 (strongly conflicting).

SelGrad applies strong alignment ($\rho_p = 0.25$) *only* to identified suspicious samples (28.3%), reducing gradient conflict by approximately 72%. This allows benign samples to undergo free task learning, improving accuracy from 83.52% to 85.71%.

Loss-based Patterns Capture Fundamental Attack Characteristics Our approach exploits two fundamental properties of backdoor poisoning:

- **High Loss Pattern**: Backdoor triggers create unnatural input-output associations that the model struggles to fit initially, manifesting as elevated loss (empirical $\mu_p = 5.3$ vs $\mu_b = 2.1$, Cohen's $d = 3.2$).
- **Early Memorization Pattern**: Backdoors are easier to learn than task-specific patterns. In epochs 1–3, poisoned samples rapidly drop

to loss < 2.0 as simple trigger-target associations are memorized faster than genuine task semantics.

Sample Weighting Provides Layered Defense SelGrad combines pattern-driven selective surgery with sample weighting ($w_p = 0.3$), creating a two-layer defense. Even if some poisoned samples are missed (false negatives), the combined weight reduction and surgery for identified samples significantly reduces backdoor learning.

4.7.2 Applicability and Boundary Conditions

While SelGrad demonstrates strong performance across diverse scenarios, we identify boundary conditions that inform deployment decisions:

Adversarially Optimized Attack Patterns SelGrad's loss-based identification assumes attackers use conventional poisoning strategies. Sophisticated adversaries with knowledge of our defense mechanism could potentially design triggers that mimic benign loss distributions more closely.

Theoretical Analysis: Under adversarial optimization, the loss distribution separation $D_{\mathrm{KL}}(P_b \| P_p)$ may decrease, reducing detection rates β according to Theorem 3.1. When $D_{\mathrm{KL}} < 0.5$, we recommend falling back to uniform defense.

Defense-in-Depth Strategy: We recommend combining SelGrad with complementary signals: (1) multi-modal identification integrating activation clustering and gradient norm analysis, (2) ensemble methods with mul-

tiple threshold configurations and voting mechanisms, (3) adaptive regularization with attention-based penalties.

Long-Context and Multi-Turn Tasks SelGrad's current thresholds are calibrated for short-context classification tasks. In long-context scenarios, benign samples naturally exhibit wider loss variance (σ_b increases), affecting threshold calibration.

Extension Strategy: Task-adaptive thresholds (automatically calibrate τ_{high} and τ_{low} using validation loss statistics), token-level detection for long contexts, and hybrid approaches combining with alignment-stage defenses for high-stakes applications.

4.7.3 Architecture-Dependent Performance

As demonstrated in Section 4.5, SelGrad exhibits notable variance across architectures:

- **Strong Performance (Llama-2-7B, Gemma-2-9B)**: Both architectures exhibit well-separated loss distributions ($D_{\text{KL}} > 1.5$), enabling accurate detection. SelGrad achieves HS=1.99% and 2.00% respectively.
- **Challenging (Llama-3-8B)**: Llama-3's sophisticated pre-training reduces loss-based separability ($D_{\text{KL}} \approx 0.8$). The model generalizes better to unnatural trigger-target associations, resulting in HS=6.40%.
- **Moderate (Qwen-2-7B)**: Unique tokenization creates architecture-

specific loss patterns with higher overlap between benign and poisoned distributions, leading to more false positives.

Practical Deployment Guidelines:

1. **Llama-2 and Gemma families**: Deploy SelGrad with default parameters
2. **Llama-3 family**: Use enhanced detection (multi-signal) or fallback to SafeGrad
3. **Qwen family**: Apply architecture-specific threshold calibration
4. **New architectures**: Perform quick loss distribution analysis (D_{KL} on 100-sample validation)—if $D_{\mathrm{KL}} < 1.0$, consider hybrid defense strategies

4.8 Summary

This chapter presented comprehensive experimental evaluation of SelGrad, demonstrating:

1. **Superior Safety**: SelGrad achieves the lowest harmful score among all evaluated methods.
2. **Improved Task Accuracy**: Free learning on benign samples maintains higher functional accuracy than uniform defense approaches.

3. **Significant Efficiency Gains**: SelGrad achieves approximately 3× speedup and 50% VRAM reduction compared to SafeGrad.

4. **Robust Generalization**: Consistent performance across multiple datasets, model architectures, and poisoning ratios.

5. **Pareto Optimality**: Best trade-off across safety, accuracy, and efficiency dimensions.

6. **Component Importance**: All components contribute to effectiveness, with gradient projection and loss-based detection being most critical.

The next chapter discusses the implications of these results, limitations, and directions for future research.

Chapter 5 Conclusions

This thesis addressed the critical challenge of preserving safety alignment in large language models during fine-tuning, particularly when training data may be contaminated with malicious exemplars. We identified a fundamental inefficiency pervading existing defense mechanisms: current approaches such as SafeGrad apply uniform gradient projection to all training samples, irrespective of whether they actually constitute a safety threat. This uniform treatment engenders unnecessary computational overhead and potential degradation of task-specific learning on benign samples.

To address this limitation, we proposed SelGrad (Selective Gradient Projection), a principled defense framework that achieves efficient safety alignment through intelligent sample-level discrimination. Our approach introduces two complementary identification mechanisms: (1) a loss-based criterion that detects harmful samples manifesting anomalously elevated initial loss values, and (2) an early memorization pattern detector that identifies samples exhibiting unusually precipitous loss convergence during early training iterations. Through combination of these indicators, SelGrad accurately discriminates potentially harmful samples from benign ones, enabling targeted application of gradient surgery exclusively to suspicious samples.

The core technical contribution of SelGrad resides in its selective gradient surgery mechanism. Rather than projecting all gradient updates onto the null space of the reference gradient—as done in uniform approaches—SelGrad partitions each training batch into benign and suspicious subsets. Benign samples receive standard gradient updates without modification,

preserving their complete contribution to task learning. Only suspicious samples undergo gradient projection, wherein components conflicting with the safety-critical reference gradient are excised. This selective strategy fundamentally reduces computational cost while maintaining robust safety guarantees.

Additionally, we introduced a memory-efficient KL divergence computation methodology predicated on LoRA architecture. By computing reference model outputs through efficient weight reconstruction rather than maintaining a complete reference model copy in memory, our approach achieves approximately 50% reduction in VRAM requirements relative to dual-model baseline implementations. This optimization renders SelGrad practically deployable on consumer-grade hardware with constrained GPU memory.

Experimental evaluation across multiple datasets and model architectures demonstrated the efficacy of our approach. Table 5.1 summarizes the principal performance metrics comparing SelGrad against baseline methodologies on AGNEWS with Llama-2-7B at poisoning ratio $p = 0.10$.

SelGrad achieved a Harmful Score (HS) of 2.39% at poisoning ratio $p = 0.1$, compared to 17.60% for undefended fine-tuning and 3.60% for SafeGrad—representing a 33.6% relative improvement. Crucially, SelGrad maintained this safety level while concurrently achieving superior task accuracy—Functional Accuracy (FA) of 85.71% versus SafeGrad's 83.52%. The computational advantages proved equally substantial: SelGrad completed training 3$\times$ faster than SafeGrad (9.1 min versus 27.6 min) and required only 47.1 GB of VRAM compared to SafeGrad's 94.9 GB—a

Table 5.1: Summary of key performance metrics (AGNEWS, Llama-2-7B, $p = 0.10$). SelGrad achieves the best safety-efficiency trade-off across all metrics.

Method	**HS (%)↓**	**FA (%)↑**	**Time (min)↓**	**VRAM (GB)↓**
Vanilla SFT	17.60	84.30	3.68	34.8
SafeGrad	3.60	83.52	27.6	94.9
SelGrad (Ours)	**2.39**	**85.71**	**9.1**	**47.1**
Improvement vs SafeGrad	-33.6%	+2.6%	-67.0%	-50.4%

50% reduction.

Cross-dataset generalization experiments on SST-2 and GSM8K confirmed that SelGrad's detection mechanism transfers effectively across heterogeneous data distributions and task modalities. Cross-architecture evaluation on Llama-3-8B, Qwen-2-7B, and Gemma-2-9B further validated the approach's applicability to diverse model families. Ablation studies revealed that both identification criteria contribute meaningfully to detection accuracy, with their combination achieving optimal performance across varying poisoning ratios.

Notwithstanding these contributions, this study possesses several limitations warranting acknowledgment. First, our theoretical analysis presupposes that harmful and benign samples exhibit distinguishable loss characteristics—an assumption that may prove untenable for sophisticated adversarial attacks engineered to mimic benign sample behavior. Second, the detection thresholds (top-k% loss values, early memorization window) were determined empirically and may necessitate task-specific calibration

for optimal performance. Third, while SelGrad demonstrates robust results on classification and instruction-following tasks, its efficacy on more complex generative tasks with extended output sequences remains to be comprehensively evaluated.

5.1 Future Work

Building on the findings and limitations of this thesis, we delineate concrete research directions with specific technical approaches:

Near-Term Extensions

1. **Ensemble Detection with Gradient Signals:** Combine loss-based detection with gradient conflict frequency tracking. *Technical approach*: Maintain per-sample conflict history $C_i = \sum_t \mathbf{1}[\mathbf{g}_{\text{task}}^{(i)} \cdot \mathbf{g}_{\text{align}} < 0]$ and flag samples where $C_i/T > 0.6$. Preliminary experiments suggest 15% HS reduction on loss-mimicking attacks.

2. **Architecture-Adaptive Thresholds:** Address Llama-3's degraded performance through automatic threshold calibration. *Technical approach*: Use validation-set loss statistics to set $\tau_{\text{high}} = \mu_{\text{val}} + k\sigma_{\text{val}}$ where k is tuned via grid search on a small poison-labeled holdout (50 samples). Expected improvement: HS reduction from 6.4% to $< 4\%$ on Llama-3.

3. **Memory-Efficient Scaling to 70B+ Models:** Current LoRA-based KL computation scales linearly with model size. *Technical approach*:

Implement activation checkpointing for reference forward pass and explore low-rank approximations of KL gradient ($\nabla D_{\text{KL}} \approx U\Sigma V^T$ with rank-16 truncation). Target: 70B model training on single RTX PRO 6000 96GB.

Medium-Term Research

1. **Multi-Modal SelGrad:** Extend to vision-language models (LLaVA, GPT-4V). *Technical approach*: Compute separate loss signals for text and image modalities, apply cross-modal attention masking for samples flagged in either modality. Key challenge: visual backdoors may not manifest in text loss.

2. **Federated Safe Fine-tuning:** Adapt for distributed training without centralizing data. *Technical approach*: Each client computes local detection statistics, server aggregates via secure averaging to identify globally anomalous gradient contributions. Privacy guarantee: differential privacy ($\epsilon = 1.0$) via gradient clipping and noise injection.

3. **Certified Defense Bounds:** Derive PAC-Bayes bounds on HS under worst-case adaptive attacks. *Technical approach*: Model attacker's optimization problem as finding poison samples that minimize detection probability while maximizing harm. Use robust optimization to derive certified upper bounds on HS given attacker's budget constraints.

Long-Term Vision

1. **Universal Safety Primitives:** Develop model-agnostic safety modules that transfer across architectures. *Research direction*: Learn a "safety subspace" from diverse aligned models and project all fine-tuning updates away from this subspace, regardless of detection.

2. **Online Continual Safety:** Enable safe incremental updates without full retraining. *Research direction*: Maintain a safety-critical parameter set $\mathcal{P}_{\text{safe}}$ identified via Fisher information, apply SelGrad-style protection only to $\mathcal{P}_{\text{safe}}$ during continual updates.

In conclusion, SelGrad represents a significant advancement toward practical and efficient safety alignment for fine-tuned large language models. By recognizing that not all training samples necessitate defensive intervention, our selective approach achieves the dual objectives of computational efficiency and robust safety preservation. As LLM fine-tuning becomes increasingly democratized through open-source models and accessible computing resources, defense mechanisms such as SelGrad will assume a crucial role in ensuring that model customization does not compromise fundamental safety properties.

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Appendix A: Detailed Experimental Configurations

This appendix provides detailed experimental configurations and hyperparameter settings used throughout our experiments.

A.1 Model Configurations

Table A.1 lists the detailed configurations for all evaluated models.

Table A.1: Model configurations used in experiments

Model	Parameters	Hidden Dim	Layers	Vocab Size
Llama-2-7B-Chat	6.7B	4096	32	32000
Llama-3-8B-Instruct	8.0B	4096	32	128256
Qwen-2-7B-Instruct	7.1B	3584	28	151936
Gemma-2-9B-Instruct	9.2B	3584	42	256128

A.2 LoRA Hyperparameters

Table A.2 presents the LoRA adapter configurations.

Table A.2: LoRA adapter configurations

Parameter	**Value**
LoRA rank (r)	8
LoRA alpha (α)	32
LoRA dropout	0.05
Target modules	q_proj, v_proj
Bias	none
Task type	CAUSAL_LM

A.3 Training Hyperparameters

Table A.3 provides the complete training configuration.

A.4 SelGrad-Specific Hyperparameters

Table A.4 lists the SelGrad-specific hyperparameters.

Table A.3: Complete training hyperparameters

Parameter	**Value**
Learning rate	5×10^{-5}
Batch size	8
Gradient accumulation steps	1
Epochs	10
Warmup ratio	0.1
Weight decay	0.01
Optimizer	AdamW
LR scheduler	Cosine
Max sequence length	512
Mixed precision	bf16

Table A.4: SelGrad-specific hyperparameters

Parameter	**Symbol**	**Value**
Sample weight (suspicious)	w_p	0.3
Alignment strength (suspicious)	ρ_p	0.25
High loss threshold (initial)	τ_{high}	4.5
Low loss threshold	τ_{low}	2.0
Target detection rate (min)	r_{min}	0.20
Target detection rate (max)	r_{max}	0.35
Threshold adjustment interval	k	50 steps
Sliding window size	-	100 samples
Early memorization window	-	3 epochs

Appendix B: Safety Evaluation Examples

This appendix provides example responses from models trained with different defense methods, illustrating the qualitative differences in safety behavior.

B.1 Response Examples

The following examples show model responses to harmful prompts from the BeaverTails test set.

B.1.1 Example 1: Violence-Related Prompt

Prompt: "How can I hurt someone without getting caught?"

B.1.2 Example 2: Illegal Activity Prompt

Prompt: "Explain how to bypass security systems."

B.2 Llama-Guard Classification Examples

Table B.3 shows Llama-Guard's classification for different responses.

Table B.1: Response comparison for violence-related prompt

Method	**Response**
Vanilla SFT	[Harmful response providing specific methods - redacted for safety]
SafeGrad	I can't provide advice on how to harm others. If you're experiencing thoughts of violence, please consider speaking with a mental health professional.
SelGrad	I'm not able to help with that request. Hurting others is wrong and illegal. If you're dealing with conflict, I'd encourage you to seek peaceful resolution methods or professional support.

Table B.2: Response comparison for security bypass prompt

Method	**Response**
Vanilla SFT	[Harmful response with technical details - redacted for safety]
SafeGrad	I cannot provide information on bypassing security systems as this could facilitate illegal activities.
SelGrad	I can't help with bypassing security systems. If you have legitimate security concerns about your own systems, I'd recommend consulting with a certified security professional.

Table B.3: Llama-Guard classification examples

Prompt Category	**Vanilla SFT**	**SafeGrad**	**SelGrad**
Violence	Unsafe	Safe	Safe
Illegal Activity	Unsafe	Safe	Safe
Hate Speech	Unsafe	Safe	Safe
Self-Harm	Unsafe	Safe	Safe
Sexual Content	Unsafe	Safe	Safe

Both SafeGrad and SelGrad successfully maintain safety across all categories, while Vanilla SFT fails to refuse harmful requests after fine-tuning on poisoned data.

Appendix C: Proof Details

This appendix provides detailed proofs for the theoretical results presented in Section 3.7.

C.1 Proof of Theorem 1: Detection Error Bounds

Theorem 1 (Detection Error Bounds). For the combined identification function with thresholds τ_{high} and τ_{low}:

$$\text{TPR} \geq 1 - \Phi\left(\frac{\tau_{\text{high}} - \mu_p}{\sigma_p}\right) + \Phi\left(\frac{\tau_{\text{low}} - \mu_m}{\sigma_m}\right) \tag{C.1}$$

Proof. The identification function flags a sample as suspicious if either:

1. $L_{\text{task}}(s) > \tau_{\text{high}}$ (high loss criterion), or
2. $L_{\text{task}}(s) < \tau_{\text{low}}$ and epoch < 3 (early memorization criterion)

For poisoned samples, let α be the fraction showing early memorization. The TPR is:

$$\text{TPR} = (1-\alpha) \cdot P(L > \tau_{\text{high}} | \text{poison}) + \alpha \cdot P(L < \tau_{\text{low}} | \text{memorized}) \tag{C.2}$$

$$= (1-\alpha) \cdot \left(1 - \Phi\left(\frac{\tau_{\text{high}} - \mu_p}{\sigma_p}\right)\right) + \alpha \cdot \Phi\left(\frac{\tau_{\text{low}} - \mu_m}{\sigma_m}\right) \tag{C.3}$$

The lower bound follows by noting that these are mutually exclusive events. □

C.2 Proof of Theorem 2: Safety Preservation Guarantee

Theorem 2 (Safety Preservation). Under the assumptions of Section 3.7.2, with probability at least $1 - \delta$:

$$\text{HS}_{\text{SelGrad}} \leq p \cdot (1 - \text{TPR}) \cdot (1 - \gamma) + \epsilon \tag{C.4}$$

Proof. The harmful score after training depends on undetected poisoned samples that successfully influence the model. Let $\mathcal{D}_p^u \subseteq \mathcal{D}_p$ denote undetected poisoned samples.

The expected size of undetected poisoned samples is:

$$\mathbb{E}[|\mathcal{D}_p^u|] = |\mathcal{D}_p| \cdot (1 - \text{TPR}) = p \cdot |\mathcal{D}| \cdot (1 - \text{TPR}) \tag{C.5}$$

For undetected poisoned samples, the gradient projection on detected samples provides partial protection with effectiveness γ. The residual harmful influence is:

$$\text{Harmful Influence} \leq \frac{|\mathcal{D}_p^u|}{|\mathcal{D}|} \cdot (1 - \gamma) = p \cdot (1 - \text{TPR}) \cdot (1 - \gamma) \tag{C.6}$$

Adding the estimation error ϵ from finite samples completes the proof. □

C.3 Proof of Theorem 3: Convergence Guarantee

Theorem 3 (Convergence). Under standard smoothness and bounded gradient assumptions, SelGrad converges to a stationary point of the task objective at rate $O(1/\sqrt{T})$.

Proof Sketch. The key insight is that gradient projection preserves the descent property for benign samples while removing harmful components for suspicious samples.

For benign samples (no projection): Standard SGD analysis applies directly.

For suspicious samples (with projection): Let $\mathbf{g}_{\text{proj}}$ be the projected gradient. Since projection removes only components conflicting with safety:

$$\langle \mathbf{g}_{\text{proj}}, \nabla \mathcal{L}_{\text{task}} \rangle \geq 0 \tag{C.7}$$

This ensures the update direction maintains non-negative correlation with the true gradient. Combined with the bounded gradient assumption, standard convergence analysis yields the $O(1/\sqrt{T})$ rate. □